PHYS 357: Honours Quantum Physics 1

W.R.YCH.

August 31st, 2026

Abstract

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1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Mathematical Background

2.1 Taylor series

Definition 1 (Taylor series)**.** *A function f that is smooth near a (it can be differentiated any number of times there) equals the power series*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n, \tag{1}$$

where $f^{(n)}$ is the nth derivative. The special case $a = 0$ is called the Maclaurin series.

Result 1 (Series used in this course)**.**

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \qquad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \tag{2}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x|<1), \qquad (1+u)^N = \sum_{n=0}^{N} \binom{N}{n} u^n \quad (N \in \mathbb{N}), \tag{3}$$

$$\ln(1+u) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} u^n}{n} = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots \quad (|u|<1). \tag{4}$$

Problem 1 (The exponential as a limit (PS2 Q9))**.** *Prove the identity $\lim_{N\to\infty} (1 + x/N)^N = e^x$ by comparing the Taylor-series expansions of both sides.*

Remark 1. *This limit is what lets a rotation be written as an exponential (Section 6.2).*

2.2 Euler's formula and trigonometric identities

Result 2 (Euler's formula)**.** *Comparing the series for* $e^{i\theta}$*,* $\cos\theta$ *and* $\sin\theta$ *gives* $e^{i\theta} = \cos\theta + i\sin\theta$*, hence*

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \tag{5}$$

These are used constantly when rotation matrices are written out.

Result 3 (Double-angle identities)**.** *Squaring* $e^{ix} = \cos x + i\sin x$ *and comparing with* $e^{2ix} = \cos 2x + i\sin 2x$*,*

$$\cos^2 x - \sin^2 x = \cos 2x, \qquad 2\sin x\cos x = \sin 2x. \tag{6}$$

These turn the half angles $\phi/2$ *in spin rotation matrices back into full angles* ϕ*.*

2.3 Bases

Definition 2 (Basis)**.** *A set of vectors is* linearly independent *if none of them can be written as a combination of the others, and it* spans *a space if every vector in the space is some combination of them. A* basis *is a set that does both, so every vector is a unique combination of the basis vectors. The number of basis vectors is the dimension of the space.*

Definition 3 (Kronecker delta)**.**

$$\delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases} \tag{7}$$

3 Stern-Gerlach Experiments

August 31 - September 9, 2026

3.1 The classical picture

3.1.1 Angular momentum

Definition 4 (Orbital angular momentum)**.** *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p},$$

where $\vec{r}$ *is the position vector and* $\vec{p} = m\vec{v}$ *is the linear momentum.*

For motion confined to the x-y plane, only the z component survives. On a circular path of radius r with speed v,

$$L_z = mvr.$$

Definition 5 (Angular momentum of a rigid body)**.** *For a rigid body*[1] *rotating about a fixed axis,*

$$L = I_0 \omega,$$

with ω the angular velocity and I_0 the moment of inertia, which measures how the mass is distributed relative to the rotation axis.

3.1.2 Magnetic moment of a circulating charge

Definition 6 (Magnetic moment)**.** *A vector $\vec{\mu}$ measuring the net circulation of electric charge. It sets the torque, and in a nonuniform field the force, that a magnet experiences in an external magnetic field.*

A point particle of mass m and charge q on a circular orbit of radius r and period T passes any given point once per period, so it acts as a steady current $I = q/T = qv/2\pi r$. Its moment is

$$\mu = \frac{IA}{c}^{2} = \frac{1}{c}\left(\frac{qv}{2\pi r}\right)\left(\pi r^2\right) = \frac{q}{2mc}\, mvr = \frac{q}{2mc}\, L_z,$$

where A is the area enclosed by the orbit.

3.1.3 Gyromagnetic ratio

Definition 7 (Gyromagnetic ratio)**.** *The constant γ relating magnetic moment to angular momentum,*

$$\vec{\mu} = \gamma \vec{L}, \qquad \gamma = \frac{q}{2mc} \text{ for a circulating charge.}$$

Remark 2. *This is purely classical: a point charge on a definite orbit, nothing quantized. It matters because measuring $\vec{\mu}$ is how we measure angular momentum, which is what the Stern-Gerlach apparatus does.*

3.2 Spin

Definition 8 (Spin)**.** *Spin $\vec{S}$ is an* intrinsic *angular momentum: a fixed property of the particle type, like charge or mass, not caused by any motion through space. A stationary electron alone in empty space still has spin. It comes in addition to any orbital $\vec{L}$, not in place of it.*

3.2.1 Intrinsic moment and the g-factor

Spin also carries a magnetic moment proportional to it, but with an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \qquad \gamma = \frac{gq}{2mc}. \tag{8}$$

[1] A rigid body is one that does not deform under the action of external forces.

[2] Gaussian (CGS) units, where a current loop has moment IA/c. In SI units it is IA.

Definition 9 (g-factor)**.** *The dimensionless number g above. It says how much stronger or weaker the moment is than the classical value ($g = 1$). It must be measured or derived from relativistic quantum theory.*

3.2.2 Magnetons

Definition 10 (Planck constant)**.** *The constant that sets the scale at which quantization matters. It relates photon energy to frequency, $E = h\nu$, and*

$$h \approx 6.626 \times 10^{-34}\ \mathrm{J\,s}, \qquad \hbar = \frac{h}{2\pi} = 1.055 \times 10^{-27}\ \mathrm{erg\,s} = 6.582 \times 10^{-16}\ \mathrm{eV\,s}.$$

Both have units of angular momentum, $[\hbar] = [L_z] = \mathrm{kg\,m^2/s}$.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $|e|\hbar/2mc$ then carries all the units and sets the natural size of the moment for a given mass.

Definition 11 (Magnetons)**.** *The natural units of magnetic moment for the electron (Bohr magneton) and for nucleons, i.e. protons and neutrons (nuclear magneton):*

$$\mu_B = \frac{|e|\hbar}{2m_e c}, \qquad \mu_N = \frac{|e|\hbar}{2m_p c} \approx \frac{\mu_B}{1836}, \tag{9}$$

since $m_p/m_e \approx 1836$.

The magnetic moments are then

$$\vec{\mu}_e = -g_e\,\mu_B\,\frac{\vec{S}}{\hbar}, \qquad \vec{\mu}_{p,n} = g_{p,n}\,\mu_N\,\frac{\vec{S}}{\hbar}, \tag{10}$$

where the minus sign for the electron comes from its negative charge.

Particle	Charge q	Mass m	g-factor	Gyromagnetic ratio γ
Electron	$-\lvert e\rvert$	m_e	$g_e \approx 2$	$-g_e\mu_B/\hbar$
Proton	$+\lvert e\rvert$	m_p	$g_p \approx 5.59$	$g_p\mu_N/\hbar$
Neutron	0	$m_n \approx m_p$	$g_n \approx -3.83$	$g_n\mu_N/\hbar$

Table 1: Charges, masses, g-factors and gyromagnetic ratios.

Remark 3. *Classically $\mu = (q/2mc)L_z$ vanishes when $q = 0$, yet the neutral neutron has a moment. The reason is that the neutron is not a point particle: it is made of three charged quarks (one up quark with charge $+\frac{2}{3}|e|$ and two down quarks with $-\frac{1}{3}|e|$ each). The charges add to zero, but the quarks' spins and motion inside the neutron each give a moment, and these moments do not cancel. So its $\vec{\mu}$ cannot come from a single circulating point charge: a first sign the classical picture fails. The same internal structure is why $g_p \approx 5.59$ is far from 2.*

Remark 4. *Since $m_p \gg m_e$, $\mu_N \ll \mu_B$, so nuclei are deflected far less than electrons in a Stern-Gerlach device.*

3.3 The Stern-Gerlach apparatus

Definition 12 (Stern-Gerlach (SG) apparatus)**.** *A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped so that the field strength changes along a chosen axis, here z (a field* gradient*). The gradient pushes each atom by an amount proportional to μ_z, and a screen records where it lands. Since $\vec{\mu} = \gamma \vec{S}$, the landing position measures S_z.*

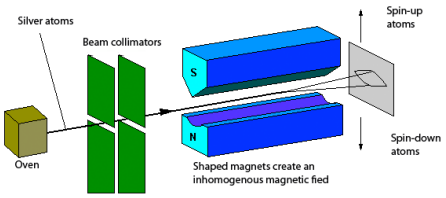


Figure 1: Illustration of the Stern-Gerlach apparatus.

3.3.1 Force in a nonuniform field

A magnetic moment in a field has potential energy $U = -\vec{\mu} \cdot \vec{B}$, so the force experienced by the particle is

$$\vec{F} = -\vec{\nabla} U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \qquad F_z = \mu_z \frac{\partial B_z}{\partial z}. \tag{11}$$

Remark 5. *A uniform field gives no net force, only a torque (a twist that turns $\vec{\mu}$ without moving the atom). That is why the magnet needs a gradient.*

Definition 13 (Deflection)**.** *The sideways displacement of an atom on the screen caused by F_z. It is proportional to μ_z, so the screen pattern is a record of the moments in the beam.*

3.3.2 Silver atoms

Silver has zero net orbital angular momentum, $\vec{L} = 0$, and its whole moment comes from one unpaired outermost (valence) electron, so the atom behaves like a single electron spin. With

$$\mu_z = -\frac{g_e |e|}{2 m_e c} S_z$$

and a gradient $\partial B_z / \partial z < 0$,

$$F_z = g_e \frac{|e|}{2 m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z.$$

Result 4. *$F_z \propto S_z$: the deflection measures the z component of spin directly.*

Problem 2 (Force proportional to spin (PS1 Q1))**.** *In a Stern-Gerlach apparatus, the vertical force on a silver atom is given by* $F_z = \mu_z \frac{\partial B_z}{\partial z}$*. Show how this force is directly proportional to the spin component* S_z*.*

3.3.3 Classical prediction versus observation

Atoms leave the oven with spins pointing in random directions, so $S_z = |\vec{S}| \cos\theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$.

Remark 6. *Classically the screen should show a continuous smear. It shows two spots.*

Result 5 (Quantization of spin)**.** *The electron's spin component, if prepared in the* $|+\hat{z}\rangle$ *state, is quantized:*

$$S_z = \pm\frac{\hbar}{2}. \tag{12}$$

3.4 Spin quantization

Now if we generalize the SG experiment for any random particles (electron, deutron, atoms, proton, neutron...), we get a specific amount of spots on the screen. The number of spots is the number of beams n, which is related to the spin quantum number s by

$$n = 2s + 1. \tag{13}$$

Definition 14 (Spin quantum number)**.** *From the amount of beams on the screen, we can determine the spin quantum number* s *of the particle.*

$$s = \frac{n-1}{2}.$$

The number s, fixed by the particle type, taking values $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$ A measurement of S_z gives

$$S_z = m\hbar.$$

Definition 15 (Magnetic quantum number)**.** *The number* m *is called the magnetic quantum number, it defines the orientation of the spin, and it takes values*

$$m = -s, -s+1, \ldots, +s.$$

Example. *To get the values of the spin of an electron we use the silver atom in the SG apparatus, since the spin of silver is defined by its lonely valence electron. We get 2 beams, so* $n = 2$*, therefore* $s = 1/2$*, and the possible values of* m *are* $m = -1/2, +1/2$*, which gives the two possible values of* S_z*:*

$$S_z = \pm\frac{\hbar}{2}.$$

Remark 7. S_z *is a component. The magnitude is* $|\vec{S}| = \hbar\sqrt{s(s+1)}$*, which for the electron (*$s = 1/2$*) is* $\hbar\sqrt{3}/2$*.*

3.5 Sequential Stern-Gerlach experiments

We write SGz for a device whose gradient points along z, and SGx for the same device rotated by 90° about the beam axis.

Definition 16 (Ket)**.** *A ket $|\cdot\rangle$ is a label for a quantum state, treated as a vector. For electron spin, the two outcomes of an SGz give*

$$|+\hat{z}\rangle \equiv |\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \qquad |-\hat{z}\rangle \equiv |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2},$$

and likewise $|\pm\hat{x}\rangle$ for $S_x = \pm\hbar/2$, and so on for any axis. We follow Townsend and use the $|\pm\hat{n}\rangle$ labels.

Definition 17 (Filter)**.** *Each device sends atoms into two beams. A filter blocks one of them.*

Remark 8. *A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation.*

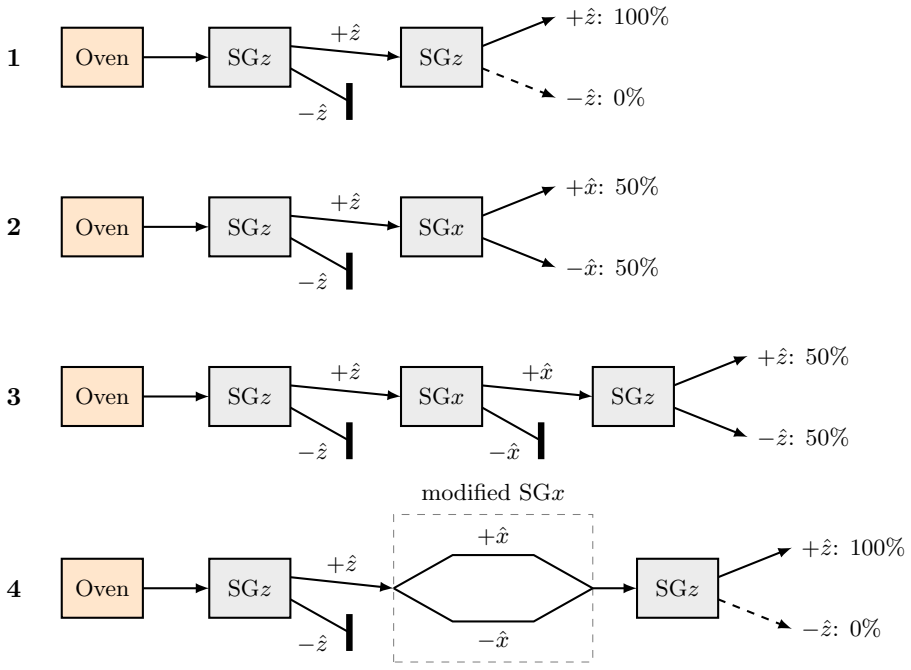


Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device. Every oven starts with N_0 atoms.

Remark 9. *The lecture notes use N_0 for the number of atoms entering the* second *device rather than leaving the oven, so their counts are twice the ones below.*

3.5.1 Experiment 1: SGz then SGz

Filter the $+\hat{z}$ beam and send it into a second SGz. Every atom comes out $+\hat{z}$. The measurement is repeatable: an atom found with $S_z = +\hbar/2$ gives the same answer again, so it left the first device in $|+\hat{z}\rangle$. We end with $N_0/2$ atoms in $|+\hat{z}\rangle$.

3.5.2 Experiment 2: SGz then SGx

Filter $+\hat{z}$ and send it into an SGx. Half come out $+\hat{x}$, half $-\hat{x}$: knowing S_z tells us nothing about S_x. We end with $N_0/4$ in each of $|\pm\hat{x}\rangle$.

Problem 3 (SGz then SGx (PS1 Q2))**.** *A beam of silver atoms is prepared in the $|+\hat{z}\rangle$ state and subsequently passes through an SGx apparatus. What are the possible measurement outcomes and their respective probabilities?*

3.5.3 Experiment 3: SGz, SGx, then SGz

Filter $+\hat{z}$, filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 even though every atom was $+\hat{z}$ at the start. Blocking $+\hat{x}$ instead gives the same result. We end with $N_0/8$ in each of $|\pm\hat{z}\rangle$.

Result 6. *No state has definite values of both S_x and S_z. Measuring S_x destroys what we knew about S_z.*

3.5.4 Experiment 4: SGz, modified SGx, then SGz

Definition 18 (Modified SG device)**.** *An SG device that splits the beam according to S_n and then recombines the two beams, so both outcomes exit at the same place without being recorded.*

Filter $+\hat{z}$, pass it through a modified SGx with neither path blocked, then measure S_z. Every atom comes out $+\hat{z}$: all $N_0/2$ end in $|+\hat{z}\rangle$.

Result 7. *The atoms cannot be going half through $+\hat{x}$ and half through $-\hat{x}$, since Experiment 3 showed each half alone gives 50/50. Instead the two paths* interfere*: as with water or light waves, contributions from the two paths add and can cancel, here cancelling the $-\hat{z}$ outcome. No definite path can be assigned to an atom. Section 4 makes this precise.*

Remark 10. *With one path blocked, a modified SG acts exactly like an ordinary SG.*

4 Quantum States

September 11-18, 2026

Throughout, an *n-level system* is one whose measurements have n possible outcomes (13).

4.1 Kets, bras and bases

Definition 19 (Bra)**.** *Every ket (Def. 16) $|\psi\rangle$ has a matching bra $\langle\psi|$, its conjugate transpose (adjoint): the ket's coefficients are complex conjugated, and the column becomes a row, its derivation is shown in Section 4.2.1.*

Definition 20 (Inner product)**.** *A bra followed by a ket gives a complex number, the inner product $\langle\phi|\psi\rangle \in \mathbb{C}$.*

Definition 21 (Orthonormal basis)**.** *The outcome states $\{|j\rangle : j = 1, \ldots, n\}$ of one measurement form an orthonormal basis of the n-dimensional state space:*

$$\langle i|j\rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \qquad \forall i, j \in \{1, \ldots, n\}, \tag{14}$$

Example. *For spin-$\frac{1}{2}$ this is $\langle \pm\hat{z}| \pm \hat{z}\rangle = 1$ and $\langle \pm\hat{z}| \mp \hat{z}\rangle = 0$, and the same holds for $\{|\pm\hat{n}\rangle\}$.*

Remark 11. *The label j counts* outcomes*, not axes. Choosing to measure S_x gives $|1\rangle = |+\hat{x}\rangle$, $|2\rangle = |-\hat{x}\rangle$; choosing S_z gives $|1\rangle = |+\hat{z}\rangle$, $|2\rangle = |-\hat{z}\rangle$. Either way $n = 2$.*

Remark 12. *A ket may also carry several labels, usually the quantum numbers that pin it down: $|s, m\rangle$. For the deuteron $|1, +1\rangle, |1, 0\rangle, |1, -1\rangle$.*

Definition 22 (Superposition)**.** *Any state is a linear combination of basis kets,*

$$|\psi\rangle = \sum_{j=1}^{n} c_j |j\rangle , \qquad c_j \in \mathbb{C}. \tag{15}$$

Example. *A state in the S_z basis is a superposition of the two outcomes, with complex coefficients $c_\pm$:*

$$\text{spin-}\tfrac{1}{2}: \ |\psi\rangle = c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle .$$

Proposition 1 (Coefficients are inner products)**.**

$$c_j = \langle j|\psi\rangle . \tag{16}$$

Proof. $\langle i|\psi\rangle = \sum_j c_j \langle i|j\rangle = \sum_j c_j \delta_{ij} = c_i$. □

Definition 23 (Projector)**.** *The outer product (a ket times a bra) $P_j = |j\rangle\langle j|$. It is an operator that keeps only the $|j\rangle$ part of a state,*

$$P_j |\psi\rangle = |j\rangle \langle j|\psi\rangle = c_j |j\rangle ,$$

and projecting twice changes nothing: $P_j^2 = |j\rangle\langle j|j\rangle\langle j| = P_j$.

Proposition 2 (Completeness relation)**.** *Because every state can be expanded in the basis, the projectors sum to the identity:*

$$\mathbb{1} = \sum_{j=1}^{n} |j\rangle \langle j| . \tag{17}$$

Example.

$$\textit{spin-}\tfrac{1}{2}: \ \mathbb{1} = |+\hat{n}\rangle \langle +\hat{n}| + |-\hat{n}\rangle \langle -\hat{n}| \ \textit{ for any axis } \hat{n}.$$

Remark 13 (Inserting the identity)**.** *Writing $\mathbb{1}$ in this form in front of a state reproduces its expansion, $|\psi\rangle = \mathbb{1}\,|\psi\rangle = \sum_j |j\rangle \langle j|\psi\rangle$. This trick is used in almost every derivation below.*

4.2 Measurement and probability

Definition 24 (Observable)**.** *A measurable quantity A. Its possible outcomes a_j each come with a state $|a_j\rangle$, and these states form an orthonormal basis.*

Example. *For spin-$\frac{1}{2}$, A can be S_x, S_y or S_z, with outcomes $\pm\hbar/2$.*

Principle 1 (Born rule)**.** *The probability that a system in $|\psi\rangle$ is found in $|\phi\rangle$ is*

$$P(\phi \,|\, \psi) = |\langle\phi|\psi\rangle|^2 \in [0, 1]. \tag{18}$$

In particular $P(A = a_j) = |\,\langle a_j|\psi\rangle\,|^2$, e.g. $P(S_z = \pm\hbar/2) = |c_\pm|^2$.

Definition 25 (Probability amplitude)**.** *The amplitude of the probability of outcome a_j is the complex number $\langle a_j|\psi\rangle$.*

Remark 14. *The Born rule has no deeper derivation: it is a postulate that agrees with every experiment. As David Mermin put it, "shut up and calculate."*

Problem 4 (Probability of S_z (PS1 Q4))**.** *Consider the quantum state $|\psi\rangle = \sqrt{\frac{1}{5}}\,|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}\,|-\hat{z}\rangle$. Calculate the probability of measuring the spin z-component to be $S_z = +\hbar/2$.*

4.2.1 Bras

Proof. Write the bra with unknown coefficients, $\langle\psi| = c'_+ \langle +\hat{z}| + c'_- \langle -\hat{z}|$. By orthonormality,

$$\langle\psi|\psi\rangle = c'_+ c_+ + c'_- c_- .$$

A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, and it is always found in one of the two beams, so this must equal $|c_+|^2 + |c_-|^2 = 1$. That forces $c'_\pm = c^*_\pm$. □

Result 8. *Going from a ket to its bra:*

$$\langle\psi| = \sum_j c_j^* \langle j| , \qquad \langle\phi|\psi\rangle = \langle\psi|\phi\rangle^* . \tag{19}$$

4.2.2 Normalization

Proposition 3 (Normalization). $\langle\psi|\psi\rangle = 1$ *is the statement that the probabilities of all outcomes add to 1.*

Proof.

$$\langle\psi|\psi\rangle = \sum_{i,j} c_i^* c_j \langle a_i|a_j\rangle = \sum_j |c_j|^2 = \sum_j P(A = a_j) = 1.$$

□

Example. *For* $|\psi\rangle = \frac{1}{\sqrt{3}}|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, $c_+ = 1/\sqrt{3}$ *and* $c_- = i\sqrt{2/3}$, *so* $P(S_z = +\hbar/2) = 1/3$ *and* $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$. *They add to 1, so the state is normalized.*

Problem 5 (Normalization condition (PS1 Q3)). *A general quantum state is expressed as the linear combination* $|\psi\rangle = c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle$. *State the mathematical normalization condition required for the probability amplitudes* c_+ *and* c_-.

4.2.3 Global and relative phase

Definition 26 (Global and relative phase). *A global phase multiplies the whole ket:* $e^{i\alpha}|\psi\rangle$ *with* $\alpha \in \mathbb{R}$.

Proposition 4. *A global phase changes no probability:* $\left|\langle\phi|e^{i\alpha}\psi\rangle\right|^2 = |e^{i\alpha}|^2 |\langle\phi|\psi\rangle|^2 = |\langle\phi|\psi\rangle|^2$ *for every* $|\phi\rangle$.

Definition 27 (Relative phase). *A relative phase sits between the terms of a superposition, like the* $e^{i\delta}$ *in* $c_+|+\hat{z}\rangle + e^{i\delta}c_-|-\hat{z}\rangle$, *and cannot be pulled out front.*

Definition 28 (Physical state). *What the system is actually like, as far as any experiment can tell. Two kets that give the same probabilities for all measurements describe the same physical state.*

Result 9. $|\psi\rangle$ *and* $e^{i\alpha}|\psi\rangle$ *are the same physical state, so global phases can be dropped. Relative phases cannot: they leave the* S_z *probabilities alone but change the* S_x *and* S_y *ones (Section 4.4).*

4.3 Expectation value and uncertainty

Definition 29 (Expectation value). *The average outcome, weighted by the probabilities:*

$$\langle A\rangle = \sum_j a_j P(A = a_j) = \sum_j a_j |\langle a_j|\psi\rangle|^2 . \tag{20}$$

Example.

$$\langle S_z\rangle = \frac{\hbar}{2}|c_+|^2 - \frac{\hbar}{2}|c_-|^2.$$

Definition 30 (Uncertainty)**.** *The standard deviation (typical spread of outcomes around the average):*

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2 , \qquad \langle A^2 \rangle = \sum_j a_j^2 \, |\langle a_j | \psi \rangle|^2 . \tag{21}$$

Remark 15. *For spin-$\frac{1}{2}$ both outcomes square to $\hbar^2/4$, so $\langle S_z^2 \rangle = \hbar^2/4$ for every state.*

Example. *For $|+\hat{x}\rangle$, Experiment 3 gives both S_z probabilities $1/2$, so $\langle S_z \rangle = 0$ and $\Delta S_z = \hbar/2$.*

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}}\,|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}\,|-\hat{z}\rangle$,*

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{3} - \frac{\hbar}{2} \cdot \frac{2}{3} = -\frac{\hbar}{6}, \qquad (\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{36} = \frac{2}{9}\hbar^2, \qquad \Delta S_z = \frac{\sqrt{2}}{3}\hbar.$$

Problem 6 (Expectation value of S_z (PS1 Q6))**.** *Calculate the exact expectation value $\langle S_z \rangle$ for the state $|\psi\rangle = \sqrt{\frac{1}{5}}\,|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}\,|-\hat{z}\rangle$.*

Problem 7 (Uncertainty in S_z (PS1 Q7))**.** *Determine the standard deviation (uncertainty) ΔS_z for the state $|\psi\rangle = \sqrt{\frac{1}{5}}\,|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}\,|-\hat{z}\rangle$ using the variance formula $(\Delta S_z)^2 = \langle S_z^2 \rangle - \langle S_z \rangle^2$.*

Problem 8 ($\langle S_z \rangle$ for $|+\hat{x}\rangle$ (PS1 Q8))**.** *The state $|+\hat{x}\rangle$ can be written as*

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} \left(|+\hat{z}\rangle + |-\hat{z}\rangle \right).$$

Show mathematically that the expectation value $\langle S_z \rangle = 0$ for this specific state.

4.3.1 Measured averages and shot noise

Remark 16. *For N measurements on identically prepared atoms, with $N_\pm$ giving $\pm\hbar/2$, the measured average is*

$$\langle S_z \rangle_N = \frac{\hbar}{2} \frac{N_+ - N_-}{N}, \tag{22}$$

which tends to $\langle S_z \rangle$ as $N \to \infty$.

Remark 17 (Shot noise)**.** *The measured average $\langle S_z \rangle_N$ scatters around the true expectation value, and its average deviation from it is*

$$|\langle S_z \rangle_N - \langle S_z \rangle| \sim \frac{\Delta S_z}{\sqrt{N}}. \tag{23}$$

This scatter is called shot noise. It comes from the randomness of individual quantum outcomes, not from a faulty apparatus, and is absent only when $\Delta S_z = 0$, i.e. when the state has a definite S_z.

4.4 Spin states along x and y

4.4.1 The state $|+\hat{x}\rangle$

Experiment 3 gives $|\langle\pm\hat{z}|+\hat{x}\rangle|^2 = \frac{1}{2}$, so both amplitudes have modulus $1/\sqrt{2}$ with their own phases:

$$|+\hat{x}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}}\left(|+\hat{z}\rangle + e^{i\delta}\,|-\hat{z}\rangle\right), \qquad \delta = \delta_- - \delta_+.$$

Dropping the global phase $e^{i\delta_+}$,

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle + e^{i\delta}\,|-\hat{z}\rangle\right).$$

Experiment 3 alone does not fix the relative phase δ.

4.4.2 Experiment 5: measuring S_y

Definition 31 (Cyclic permutation)**.** *Relabelling the axes $x \to y \to z \to x$ (or the reverse). It is a rotation of the coordinate system, so no physics changes.*

Experiment 5 is Experiment 3 cyclically permuted: filter $+\hat{z}$, then measure S_y. By symmetry it also splits 50/50, so every cross-axis overlap has modulus squared $\frac{1}{2}$:

$$|\langle+\hat{x}|+\hat{z}\rangle|^2 = |\langle+\hat{y}|+\hat{z}\rangle|^2 = |\langle-\hat{x}|+\hat{y}\rangle|^2 = \frac{1}{2}.$$

The same argument as for $|+\hat{x}\rangle$ gives, after dropping global phases,

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm e^{i\delta}\,|-\hat{z}\rangle\right), \qquad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm e^{i\gamma}\,|-\hat{z}\rangle\right).$$

4.4.3 Fixing the phases

Using orthonormality of the $\hat{z}$ basis,

$$|\langle+\hat{x}|+\hat{y}\rangle|^2 = \left|\frac{1}{2}\left(1+e^{i(\gamma-\delta)}\right)\right|^2 = \frac{1}{2}\left(1+\cos(\gamma-\delta)\right).$$

Setting this to $\frac{1}{2}$ forces $\gamma - \delta = \pm\pi/2$. The rest is convention.

Result 10 (Phase convention)**.** *With $\delta = 0$ and $\gamma = +\pi/2$,*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm |-\hat{z}\rangle\right), \qquad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm i\,|-\hat{z}\rangle\right), \tag{24}$$

$$\langle\pm\hat{x}| = \frac{1}{\sqrt{2}}\left(\langle+\hat{z}| \pm \langle-\hat{z}|\right), \qquad \langle\pm\hat{y}| = \frac{1}{\sqrt{2}}\left(\langle+\hat{z}| \mp i\,\langle-\hat{z}|\right). \tag{25}$$

Remark 18. *Complex coefficients are required: with real ones the relative phases could only be 0 or π, and $\gamma - \delta = \pm\pi/2$ would be impossible.*

Problem 9 (Probability of S_x (PS1 Q5))**.** *For the state* $|\psi\rangle = \sqrt{\frac{1}{5}}\,|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}\,|-\hat{z}\rangle$*, calculate the probability of measuring* $S_x = +\hbar/2$*.*

Problem 10 (The overlap $\langle +\hat{y}| + \hat{x}\rangle$ (PS1 Q9))**.** *Given the state vectors* $|\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm i\,|-\hat{z}\rangle\right)$*, calculate the inner product* $\langle +\hat{y}| + \hat{x}\rangle$*.*

Problem 11 (Probability of S_y from $|+\hat{x}\rangle$ (PS1 Q10))**.** *Using the inner product from Problem 10, determine the probability of measuring* $S_y = +\hbar/2$ *when a system is initially prepared in the* $|+\hat{x}\rangle$ *state.*

Example. *For* $|\psi\rangle = \sqrt{\frac{1}{3}}\,|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}\,|-\hat{z}\rangle$*, compute* $\langle S_y\rangle$*. Using* $\langle \pm\hat{y}| + \hat{z}\rangle = \frac{1}{\sqrt{2}}$ *and* $\langle \pm\hat{y}| - \hat{z}\rangle = \mp\frac{i}{\sqrt{2}}$*,*

$$\langle \pm\hat{y}|\psi\rangle = \frac{1}{\sqrt{2}}\left(\sqrt{\tfrac{1}{3}} \pm \sqrt{\tfrac{2}{3}}\right), \qquad P\big(S_y = \pm\tfrac{\hbar}{2}\big) = \frac{1}{2} \pm \frac{\sqrt{2}}{3},$$

which add to 1. Then

$$\langle S_y\rangle = \frac{\hbar}{2}\left[\left(\frac{1}{2} + \frac{\sqrt{2}}{3}\right) - \left(\frac{1}{2} - \frac{\sqrt{2}}{3}\right)\right] = \frac{\sqrt{2}}{3}\hbar.$$

The factor i *did nothing for* S_z*, but as a relative phase it is exactly what tilts the spin toward* $+\hat{y}$*.*

5 Matrix Mechanics

September 21-25, 2026

Chapter 2

5.1 States as column vectors

Definition 32 (Representation of a state)**.** *In a basis, a ket is represented by the column of its coefficients and its bra by the conjugate row (the adjoint,* †*). In the* S_z *basis:*

$$|\psi\rangle \xrightarrow{S_z} \psi = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \qquad \langle\psi| \xrightarrow{S_z} \psi^\dagger = \begin{pmatrix} c_+^* & c_-^* \end{pmatrix}, \qquad c_\pm = \langle \pm\hat{z}|\psi\rangle\,. \tag{26}$$

Remark 19. *Write* $|\psi\rangle \xrightarrow{S_z} \psi$*, not* $|\psi\rangle = \psi$*: the ket is basis-independent, the column is not.*

Example.

$$|+\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

$$|\pm\hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \qquad |\pm\hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Problem 12 (Other bases)**.** *In the S_x basis, $|\psi\rangle \xrightarrow{S_x} \begin{pmatrix}\langle +\hat{x}|\psi\rangle \\ \langle -\hat{x}|\psi\rangle\end{pmatrix}$, and similarly for S_y. Write the column representing $|+\hat{z}\rangle$ in the S_z, S_x and S_y bases.*

Proposition 5 (Inner product as a matrix product)**.** *If $|\phi\rangle \xrightarrow{S_z} \phi$ and $|\psi\rangle \xrightarrow{S_z} \psi$, then $\langle\phi|\psi\rangle = \phi^\dagger\psi$ (row times column), so $P(\phi\,|\,\psi) = \left|\phi^\dagger\psi\right|^2$ and $\langle\psi|\psi\rangle = \psi^\dagger\psi$.*

Remark 20. *For a state given in another basis, replace each basis ket by its S_z column and add. For $|\psi\rangle = a\,|+\hat{y}\rangle + b\,|-\hat{y}\rangle$,*

$$\psi \xrightarrow{S_z} \frac{a}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix} + \frac{b}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}a+b\\ i(a-b)\end{pmatrix}.$$

Checking $\psi^\dagger\psi = 1$ before going further catches most mistakes. A cleaner systematic method is in Section 5.4.

Problem 13 (A probability by matrix mechanics (PS2 Q1))**.** *Using matrix mechanics, find $P(S_x = +\hbar/2)$ for the state $|\psi\rangle = \sqrt{\frac{1}{5}}\,|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}\,|-\hat{z}\rangle$.*

Problem 14 (Probability of S_x from a $|\pm\hat{y}\rangle$ state (PS2 Q2))**.** *For the state $|\psi\rangle = \frac{1}{\sqrt{6}}\,|+\hat{y}\rangle - i\frac{\sqrt{5}}{\sqrt{6}}\,|-\hat{y}\rangle$, use the fact that $|\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm i\,|-\hat{z}\rangle\right)$ and $|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm |-\hat{z}\rangle\right)$ to find the probability that a measurement of S_x gives $+\hbar/2$ using matrix mechanics.*

5.2 Operators as matrices

Definition 33 (Linear operator)**.** *A map A from kets to kets that respects superpositions, $A\left(a\,|\psi\rangle + b\,|\phi\rangle\right) = aA\,|\psi\rangle + bA\,|\phi\rangle$. Every operator in this course is linear.*

Definition 34 (Matrix of an operator)**.** *In a basis $\{|j\rangle\}$, A is represented by the matrix of* matrix elements

$$A_{ij} = \langle i|\,A\,|j\rangle\,. \tag{27}$$

Proposition 6 (Operator acting on a state)**.** *$A\,|\psi\rangle$ is represented by the matrix times the column, $A\psi$.*

Proof. The kth entry of the column for $A\,|\psi\rangle$ is $\langle k|\,A\,|\psi\rangle$. Inserting the identity,

$$\langle k|\,A\,|\psi\rangle = \sum_j \langle k|\,A\,|j\rangle\,\langle j|\psi\rangle = \sum_j A_{kj}\,c_j,$$

which is row k of A times the column ψ. □

Example. *The projectors and identity in the S_z basis:*

$$P_{+\hat{z}} \xrightarrow{S_z} \begin{pmatrix}1&0\\0&0\end{pmatrix}, \qquad P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix}0&0\\0&1\end{pmatrix}, \qquad \mathbb{1} \xrightarrow{S_z} \begin{pmatrix}1&0\\0&1\end{pmatrix},$$

and indeed $P_{+\hat{z}}\psi = (c_+, 0)^T$.

Definition 35 (Adjoint)**.** *The adjoint $A^\dagger$ is the conjugate transpose: $(A^\dagger)_{ij} = A^*_{ji}$. In operator form, writing $A = \sum_{i,j} A_{ij} |i\rangle \langle j|$,*

$$A^\dagger = \sum_{i,j} A^*_{ij} |j\rangle \langle i| . \tag{28}$$

For real entries it is just the transpose.

Proposition 7 (Bra of a transformed state)**.** *If $|\psi'\rangle = A|\psi\rangle$, then $\langle\psi'| = \langle\psi| A^\dagger$.*

Proof. The column of $|\psi'\rangle$ is $A\psi$, and its bra is the conjugate transpose, $(A\psi)^\dagger = \psi^\dagger A^\dagger$. □

Proposition 8 (Adjoint of a product)**.**

$$(AB)^\dagger = B^\dagger A^\dagger. \tag{29}$$

The order reverses; $A^\dagger B^\dagger$ is a different matrix in general.

Problem 15 (Adjoint of a product, explicitly (PS2 Q10))**.** *Show that $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$ for two operators $\hat{A}$ and $\hat{B}$ having matrix representations*

$$\hat{A} \to \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad \hat{B} \to \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}.$$

5.3 Observables as operators

Definition 36 (Eigenstate and eigenvalue)**.** *$|a\rangle$ is an eigenstate of A with eigenvalue a if*

$$A|a\rangle = a|a\rangle , \tag{30}$$

the familiar $Mv = \lambda v$ in matrix form. A general operator turns a state into a different state; eigenstates come back as themselves times a number. A basis made of eigenstates of A is called an eigenbasis *of A.*

Remark 21 (Eigenstate vs eigenvector)**.** *Same thing in two languages. "Eigenstate" is the physics word for the ket, e.g. $|+\hat{x}\rangle$. "Eigenvector" is the linear-algebra word, usually for its column in some basis, e.g. $\frac{1}{\sqrt{2}}(1,1)^T$ in the S_z basis.*

Proposition 9 (Operator of an observable)**.** *For an observable with outcomes a_j and states $|a_j\rangle$, define*

$$\hat{A} = \sum_j a_j |a_j\rangle \langle a_j| . \tag{31}$$

Then $\hat{A}|a_j\rangle = a_j|a_j\rangle$ (the outcomes are its eigenvalues, the outcome states its eigenstates), and

$$\langle A\rangle = \langle\psi|\hat{A}|\psi\rangle , \qquad \langle A^2\rangle = \langle\psi|\hat{A}^2|\psi\rangle . \tag{32}$$

Proof. $\hat{A}\,|a_k\rangle = \sum_j a_j\,|a_j\rangle\,\delta_{jk} = a_k\,|a_k\rangle$, and $\langle\psi|\,\hat{A}\,|\psi\rangle = \sum_j a_j\,\langle\psi|a_j\rangle\,\langle a_j|\psi\rangle = \sum_j a_j\,|\langle a_j|\psi\rangle|^2 = \langle A\rangle$. Likewise $\hat{A}^2 = \sum_j a_j^2\,|a_j\rangle\,\langle a_j|$. □

Proposition 10 (An operator is diagonal in its own eigenbasis)**.**

$$\langle a_i|\,\hat{A}\,|a_j\rangle = a_j\delta_{ij}, \qquad \hat{A} \xrightarrow{|a_j\rangle} D = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix}. \tag{33}$$

5.3.1 Spin operators and Pauli matrices

Definition 37 (Spin operators)**.** *Applying the operator of an observable to spin, for* $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$*,*

$$\hat{S}_n = \frac{\hbar}{2}\left(|+\hat{n}\rangle\,\langle+\hat{n}| - |-\hat{n}\rangle\,\langle-\hat{n}|\right), \qquad \hat{S}_n\,|\pm\hat{n}\rangle = \pm\frac{\hbar}{2}\,|\pm\hat{n}\rangle\,. \tag{34}$$

Example. *Each spin operator in its own eigenbasis, using* $\hat{S}_n\,|\pm\hat{n}\rangle = \pm\frac{\hbar}{2}\,|\pm\hat{n}\rangle$ *and orthonormality:*

$$\langle\pm\hat{n}|\,\hat{S}_n\,|\pm\hat{n}\rangle = \pm\frac{\hbar}{2}, \qquad \langle\pm\hat{n}|\,\hat{S}_n\,|\mp\hat{n}\rangle = \mp\frac{\hbar}{2}\,\langle\pm\hat{n}|\mp\hat{n}\rangle = 0,$$

so

$$\hat{S}_x \xrightarrow{S_x} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \qquad \hat{S}_y \xrightarrow{S_y} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \qquad \hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

In the S_z basis the matrix elements of $\hat{S}_n$ are

$$\langle i|\,\hat{S}_n\,|j\rangle = \frac{\hbar}{2}\Big(\,\langle i| + \hat{n}\rangle\,\langle+\hat{n}|j\rangle - \langle i| - \hat{n}\rangle\,\langle-\hat{n}|j\rangle\,\Big), \qquad i,j \in \{+\hat{z}, -\hat{z}\}.$$

Problem 16 (The matrix of $\hat{S}_z$ (PS2 Q3))**.** *Find the matrix representation of* $\hat{S}_z$ *in the* S_z *basis.*

Problem 17 (The matrix of $\hat{S}_x$ (PS2 Q4))**.** *Find the matrix representation of* $\hat{S}_x$ *in the* S_z *basis.*

Problem 18 (The matrix of $\hat{S}_y$ (PS2 Q5))**.** *Find the matrix representation of* $\hat{S}_y$ *in the* S_z *basis.*

Remark 22 (Shortcut)**.** *Write each projector as a column times its conjugate row, e.g.*

$$|\pm\hat{x}\rangle\,\langle\pm\hat{x}| \xrightarrow{S_z} \frac{1}{2}\begin{pmatrix}1\\ \pm1\end{pmatrix}\begin{pmatrix}1 & \pm1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & \pm1\\ \pm1 & 1\end{pmatrix},$$

then subtract and multiply by $\hbar/2$*.*

Definition 38 (Pauli matrices)**.** *The results, all in the S_z basis:*

$$\hat{S}_j = \frac{\hbar}{2}\sigma_j, \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{35}$$

Each equals its own adjoint and squares to $\mathbb{1}$, so $\hat{S}_j^2 = \frac{\hbar^2}{4}\mathbb{1}$. Collected into the vector of Pauli matrices $\vec{\sigma} = \sigma_x\hat{x} + \sigma_y\hat{y} + \sigma_z\hat{z}$, this reads $\vec{S} \xrightarrow{S_z} \frac{\hbar}{2}\vec{\sigma}$.

Problem 19 (Expectation value and uncertainty by matrix mechanics (PS2 Q6))**.** *For $|\psi\rangle = \sqrt{\frac{1}{5}}\,|+\hat{y}\rangle + i\sqrt{\frac{4}{5}}\,|-\hat{y}\rangle$, find $\langle S_z\rangle$ and ΔS_z using matrix mechanics.*

5.3.2 Hermitian operators

Definition 39 (Hermitian operator)**.** *A is Hermitian if it equals its own adjoint, $A = A^\dagger$ (e.g. the Pauli matrices).*

Principle 2. *Every observable is represented by a Hermitian operator.*

Proposition 11. *The eigenvalues of a Hermitian operator are real, so measurement outcomes are real.*

Proof. If $A\,|a\rangle = a\,|a\rangle$, the bra is $\langle a|\,A^\dagger = a^*\,\langle a|$, so $a\,\langle a|a\rangle = \langle a|\,A\,|a\rangle = \langle a|\,A^\dagger\,|a\rangle = a^*\,\langle a|a\rangle$, and $a = a^*$. □

5.4 Change of basis

Take an n-level system with two orthonormal bases, $\{|j\rangle\}$ and $\{|\alpha\rangle\}$, $j, \alpha = 1, \ldots, n$. Within each basis the overlaps are Kronecker deltas, but across bases they are not: $\langle +\hat{x}| + \hat{z}\rangle = \frac{1}{\sqrt{2}}$ is neither 0 nor 1. The same ket then has two columns,

$$|\psi\rangle \xrightarrow{|j\rangle} \psi^j = \begin{pmatrix} \langle 1|\psi\rangle \\ \vdots \\ \langle n|\psi\rangle \end{pmatrix}, \qquad |\psi\rangle \xrightarrow{|\alpha\rangle} \psi^\alpha = \begin{pmatrix} \langle \alpha{=}1|\psi\rangle \\ \vdots \\ \langle \alpha{=}n|\psi\rangle \end{pmatrix}.$$

Inserting $\mathbb{1} = \sum_\alpha |\alpha\rangle\,\langle\alpha|$ into each $|j\rangle$ amplitude,

$$\langle j|\psi\rangle = \sum_\alpha \langle j|\alpha\rangle\,\langle\alpha|\psi\rangle\,.$$

Result 11 (Change-of-basis matrix)**.**

$$\psi^j = U\psi^\alpha, \qquad \psi^\alpha = U^\dagger\psi^j, \qquad U_{j\alpha} = \langle j|\alpha\rangle\,. \tag{36}$$

The columns of U are the new basis kets $|\alpha\rangle$ written in the old basis $\{|j\rangle\}$.

Definition 40 (Unitary operator)**.** *U is unitary if $U^\dagger U = UU^\dagger = \mathbb{1}$, i.e. its inverse is its adjoint.*

Theorem 1. *The change-of-basis matrix U is unitary.*

Proof. With $(U^\dagger)_{\alpha j} = \langle j|\alpha\rangle^* = \langle \alpha|j\rangle$ and completeness of $\{|j\rangle\}$,

$$(U^\dagger U)_{\alpha\beta} = \sum_j \langle \alpha|j\rangle \langle j|\beta\rangle = \langle \alpha|\beta\rangle = \delta_{\alpha\beta}.$$

Completeness of $\{|\alpha\rangle\}$ gives $UU^\dagger = \mathbb{1}$ the same way. □

Remark 23. *So normalization does not depend on the basis: $\psi^{j\dagger}\psi^j = \psi^{\alpha\dagger}U^\dagger U\psi^\alpha = \psi^{\alpha\dagger}\psi^\alpha$.*

Remark 24. *If a unitary U maps $|\psi\rangle$ to $c\,|\phi\rangle$ with $|\phi\rangle$ normalized, then $1 = \langle\psi|\,U^\dagger U\,|\psi\rangle = |c|^2$, so $c = e^{i\varphi}$. In particular, eigenvalues of a unitary are pure phases, and U leaves a physical state unchanged only when that state is an eigenstate of U.*

Example (Between S_x and S_z)**.** *With $|j\rangle = |\pm\hat{z}\rangle$ and $|\alpha\rangle = |\pm\hat{x}\rangle$, the columns of U are the S_z columns of $|\pm\hat{x}\rangle$:*

$$U = \begin{pmatrix} \langle +\hat{z}|+\hat{x}\rangle & \langle +\hat{z}|-\hat{x}\rangle \\ \langle -\hat{z}|+\hat{x}\rangle & \langle -\hat{z}|-\hat{x}\rangle \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{37}$$

The two directions are

$$S_x \to S_z:\ \psi^z = U\psi^x, \qquad S_z \to S_x:\ \psi^x = U^\dagger\psi^z = U\psi^z,$$

since U is real and symmetric, so $U^\dagger = U$: the same matrix goes both ways ($U^2 = \mathbb{1}$). Bill calls U "$R\left(\frac{\pi}{2}\hat{y}\right)$", but it is not the actual rotation matrix (see the warning in Section 6.4).

Remark 25. *Between S_y and S_z the matrix has columns $\frac{1}{\sqrt{2}}(1, \pm i)^T$,*

$$U_y = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \qquad \psi^z = U_y\psi^y, \qquad \psi^y = U_y^\dagger\psi^z,$$

and here $U_y^\dagger \neq U_y$, so the reverse direction needs the actual adjoint.

Problem 20 (Expectation value from the S_x basis)**.** *For $|\psi\rangle = \frac{1}{\sqrt{3}}\,|+\hat{x}\rangle + i\sqrt{\frac{2}{3}}\,|-\hat{x}\rangle$, find $\langle S_z\rangle = \psi^{z\dagger}\hat{S}_z\psi^z$ using matrix mechanics.*

5.4.1 Operators in another basis

Let $\{|j\rangle\} = \{|a_j\rangle\}$ be the eigenbasis of $\hat{A}$, so $\hat{A} \xrightarrow{|a_j\rangle} D$ (diagonal), and $U_{j\alpha} = \langle a_j|\alpha\rangle$.

Proposition 12 (Similarity transformation)**.**

$$\hat{A} \xrightarrow{|\alpha\rangle} A^\alpha = U^\dagger D\,U. \tag{38}$$

Rewriting the same operator in a new basis this way is called a similarity transformation.

Proof. Insert the identity twice:

$$\langle\alpha|\,\hat{A}\,|\beta\rangle = \sum_{i,j}\langle\alpha|a_i\rangle\,\langle a_i|\,\hat{A}\,|a_j\rangle\,\langle a_j|\beta\rangle = \sum_{i,j}(U^\dagger)_{\alpha i}D_{ij}U_{j\beta} = (U^\dagger D\,U)_{\alpha\beta}.$$

□

Result 12 (Expectation value in any basis)**.**

$$\langle A\rangle = \psi^{j\dagger}D\,\psi^j = \psi^{\alpha\dagger}\left[U^\dagger D\,U\right]\psi^\alpha. \tag{39}$$

Either move the state to the eigenbasis, or move the operator to the state's basis. Both give the same number.

Problem 21 ($\hat{S}_z$ in the S_x basis)**.** *Using U from Eq. (37), find the matrix of $\hat{S}_z$ in the S_x basis, and use it to redo the problem "Expectation value from the S_x basis" without changing the state's basis.*

6 Rotations

September 21-28, 2026

6.1 The rotation operator

Definition 41 (Rotation operator)**.** *$R(\theta\hat{n})$ is the linear operator that rotates a state by angle θ about the axis $\hat{n}$: $|\psi'\rangle = R(\theta\hat{n})\,|\psi\rangle$.*

Example. *$R\left(\frac{\pi}{2}\hat{y}\right)$ turns $\hat{z}$ into $\hat{x}$, so it takes $|\pm\hat{z}\rangle$ to $|\pm\hat{x}\rangle$ up to phase, and by linearity*

$$R\left(\tfrac{\pi}{2}\hat{y}\right)\left(c_+\,|+\hat{z}\rangle + c_-\,|-\hat{z}\rangle\right) = c_+\,|+\hat{x}\rangle + c_-\,|-\hat{x}\rangle\,.$$

Up to a phase on one column, its S_z matrix is the change-of-basis matrix U of Eq. (37) (exact form in Section 6.4).

Remark 26 (Why the bra carries $R^\dagger$)**.** *By the bra-of-a-transformed-state proposition, $\langle\psi'| = \langle\psi|\,R^\dagger$. Using $\langle\psi|\,R$ instead fails: with $|+\hat{x}\rangle = R\left(\frac{\pi}{2}\hat{y}\right)|+\hat{z}\rangle$ it would give*

$$\langle+\hat{x}|+\hat{x}\rangle = \langle+\hat{z}|\,R\left(\tfrac{\pi}{2}\hat{y}\right)R\left(\tfrac{\pi}{2}\hat{y}\right)|+\hat{z}\rangle = \langle+\hat{z}|\,R(\pi\hat{y})\,|+\hat{z}\rangle \propto \langle+\hat{z}|-\hat{z}\rangle = 0,$$

since a π rotation about $\hat{y}$ flips $+\hat{z}$ to $-\hat{z}$. That contradicts $\langle+\hat{x}|+\hat{x}\rangle = 1$.

Theorem 2. *Rotation operators are unitary: $R^\dagger R = RR^\dagger = \mathbb{1}$.*

Proof. A rotation must preserve normalization: $\langle\psi'|\psi'\rangle = \langle\psi|\,R^\dagger R\,|\psi\rangle = 1$ for every $|\psi\rangle$, which forces $R^\dagger R = \mathbb{1}$. □

6.2 Generators of rotations

Rotations about the same axis add, $R(\theta_1\hat{z})R(\theta_2\hat{z}) = R\big((\theta_1+\theta_2)\hat{z}\big)$, so a finite rotation is $N \gg 1$ small ones,

$$R(\theta\hat{z}) = \left[R\left(\tfrac{\theta}{N}\hat{z}\right)\right]^N. \tag{40}$$

Taylor expanding one small step about the identity and naming the first-order operator $-iJ_z/\hbar$,

$$R\left(\tfrac{\theta}{N}\hat{z}\right) = \mathbb{1} - \frac{i}{\hbar}J_z\frac{\theta}{N} + \mathcal{O}\left(\frac{1}{N^2}\right). \tag{41}$$

Result 13. *By the problem "The exponential as a limit",*

$$R(\theta\hat{z}) = \lim_{N\to\infty}\left(\mathbb{1} - \frac{i}{\hbar}J_z\frac{\theta}{N}\right)^N = e^{-iJ_z\theta/\hbar}, \tag{42}$$

and likewise $R(\theta\hat{n}) = e^{-iJ_n\theta/\hbar}$ for any axis.

Definition 42 (Generator of rotations)**.** *The operator J_z above: the thing that, exponentiated, produces rotations about $\hat{z}$. It has units of $\hbar$, i.e. of angular momentum ($[\hbar] = [\vec{r}\times\vec{p}]$), so $\vec{J}$ is identified as the angular momentum operator. For spin-$\frac{1}{2}$, $\vec{J} = \vec{S}$ and $R(\theta\hat{n}) = e^{-i\theta\hat{S}_n/\hbar}$.*

Proposition 13. *The generator J_z is Hermitian (Def. 39).*

Proof. To first order in θ, $R = \mathbb{1} - i\theta J_z/\hbar$ and $R^\dagger = \mathbb{1} + i\theta J_z^\dagger/\hbar$. Unitarity gives

$$\mathbb{1} = R^\dagger R = \mathbb{1} + \frac{i\theta}{\hbar}\left(J_z^\dagger - J_z\right) + \mathcal{O}(\theta^2),$$

so $J_z^\dagger = J_z$. □

Remark 27. *So J_z qualifies as an observable (Principle 2), consistent with identifying it as angular momentum.*

6.3 Functions of an operator

Proposition 14 (Function of an operator on an eigenstate)**.** *If $\hat{A}\,|a\rangle = a\,|a\rangle$ and f has a Taylor series, then*

$$f(\hat{A})\,|a\rangle = f(a)\,|a\rangle. \tag{43}$$

Proof. $f(\hat{A})$ is defined by its Taylor series, and $\hat{A}^n\,|a\rangle = a^n\,|a\rangle$, so term by term

$$f(\hat{A})\,|a\rangle = \sum_n \frac{f^{(n)}(0)}{n!}\hat{A}^n\,|a\rangle = \sum_n \frac{f^{(n)}(0)}{n!}a^n\,|a\rangle = f(a)\,|a\rangle.$$

□

Proposition 15 (Spectral form)**.** *Writing an operator as a sum of its eigenvalues times the projectors onto its eigenstates is its spectral form. Inserting* $\mathbb{1} = \sum_j |a_j\rangle \langle a_j|$ *and using Proposition 14,*

$$f(\hat{A}) = \sum_j f(a_j) |a_j\rangle \langle a_j| . \tag{44}$$

With $f(x) = x$ *this is the observable operator* $\hat{A} = \sum_j a_j |a_j\rangle \langle a_j|$.

6.3.1 Rotations about $\hat{z}$

Result 14. *With* $f(x) = e^{-i\phi x/\hbar}$ *and* $\hat{S}_z |\pm\hat{z}\rangle = \pm\frac{\hbar}{2} |\pm\hat{z}\rangle$,

$$R(\phi\hat{z}) |\pm\hat{z}\rangle = e^{\mp i\phi/2} |\pm\hat{z}\rangle , \qquad R(\phi\hat{z}) \xrightarrow{S_z} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} . \tag{45}$$

This makes sense: a rotation about $\hat{z}$ *cannot change* S_z*, so it can only attach phases to the* S_z *states. In other bases the matrix is not diagonal.*

Remark 28. *A full turn,* $\phi = 2\pi$*, gives* $R = -\mathbb{1}$*: a spin-*$\frac{1}{2}$ *state needs* 4π *to return to itself. The sign is a global phase, so it is invisible on its own. Its consequences for many particles are in Section 7.7.*

Problem 22 ($R(\phi\hat{z})$ in the S_x basis (PS2 Q7))**.** *Find the matrix representation of* $R(\phi\hat{z})$ *in the* S_x *basis.*

Problem 23 ($R(\phi\hat{z})$ in the S_y basis (PS2 Q8))**.** *Find the matrix representation of* $R(\phi\hat{z})$ *in the* S_y *basis.*

6.4 Rotations about any axis

Proposition 16 (Spectral form of a rotation)**.** *For* $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$,

$$R(\phi\hat{n}) = e^{-i\phi/2} |+\hat{n}\rangle \langle +\hat{n}| + e^{i\phi/2} |-\hat{n}\rangle \langle -\hat{n}| . \tag{46}$$

Proof. Spectral form with $f(x) = e^{-i\phi x/\hbar}$, $\hat{A} = \hat{S}_n$, eigenvalues $\pm\hbar/2$. □

Remark 29 (Recipe)**.** *In any basis* $\{|i\rangle\}$ *this reads* $VDV^\dagger$*, with* $D = \operatorname{diag}\left(e^{-i\phi/2}, e^{i\phi/2}\right)$ *and* V *the matrix whose columns are* $|\pm\hat{n}\rangle$ *written in* $\{|i\rangle\}$*. It is the similarity transformation with the roles of the two bases swapped.*

Proposition 17 ($R(\phi\hat{x})$ in the S_z basis)**.**

$$R(\phi\hat{x}) \xrightarrow{S_z} \begin{pmatrix} \cos\frac{\phi}{2} & -i\sin\frac{\phi}{2} \\ -i\sin\frac{\phi}{2} & \cos\frac{\phi}{2} \end{pmatrix} . \tag{47}$$

Proof. Here $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = V^\dagger$, so

$$R(\phi\hat{x}) \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} e^{-i\phi/2} + e^{i\phi/2} & e^{-i\phi/2} - e^{i\phi/2} \\ e^{-i\phi/2} - e^{i\phi/2} & e^{-i\phi/2} + e^{i\phi/2} \end{pmatrix} ,$$

and Euler's formula gives $2\cos\frac{\phi}{2}$ and $-2i\sin\frac{\phi}{2}$. □

Remark 30. *This is the same matrix as $R(\phi\hat{z})$ in the S_x basis. It is also the same as $R(\phi\hat{z})$ in the S_y basis (Problem 23): with $|\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm i\,|-\hat{z}\rangle\right)$,*

$$\langle\pm\hat{y}|\,R(\phi\hat{z})\,|\pm\hat{y}\rangle = \cos\frac{\phi}{2}, \qquad \langle\pm\hat{y}|\,R(\phi\hat{z})\,|\mp\hat{y}\rangle = -i\sin\frac{\phi}{2},$$

so in both bases

$$R(\phi\hat{z}) \longrightarrow \begin{pmatrix} \cos\frac{\phi}{2} & -i\sin\frac{\phi}{2} \\ -i\sin\frac{\phi}{2} & \cos\frac{\phi}{2} \end{pmatrix}.$$

Proposition 18 ($R(\phi\hat{y})$ in the S_z basis)**.**

$$R(\phi\hat{y}) \xrightarrow{S_z} \begin{pmatrix} \cos\frac{\phi}{2} & -\sin\frac{\phi}{2} \\ \sin\frac{\phi}{2} & \cos\frac{\phi}{2} \end{pmatrix}. \tag{48}$$

Proof. Now $V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$ and $V^\dagger = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \neq V$. Then

$$R(\phi\hat{y}) \xrightarrow{S_z} \frac{1}{2}\begin{pmatrix} e^{-i\phi/2} + e^{i\phi/2} & -i\left(e^{-i\phi/2} - e^{i\phi/2}\right) \\ i\left(e^{-i\phi/2} - e^{i\phi/2}\right) & e^{-i\phi/2} + e^{i\phi/2} \end{pmatrix},$$

and with $e^{-i\phi/2} - e^{i\phi/2} = -2i\sin\frac{\phi}{2}$ the off-diagonal entries are $\mp\sin\frac{\phi}{2}$. □

Remark 31. *This matrix is real, so $R(\phi\hat{y})$ never introduces factors of i into an S_z column. At $\phi = \pi$ it gives $R(\pi\hat{y})\,|+\hat{z}\rangle = |-\hat{z}\rangle$, the flip used above to show bras carry $R^\dagger$.*

Remark 32 (WARNING: rotation versus change of basis)**.** *Plugging $\phi = \pi/2$ into this proposition does* not *give Bill's change-of-basis matrix:*

$$R\left(\tfrac{\pi}{2}\hat{y}\right) \xrightarrow{S_z} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \neq U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The true rotation sends $|-\hat{z}\rangle$ to $-\,|-\hat{x}\rangle$, not $|-\hat{x}\rangle$, so its second column is $-\,|-\hat{x}\rangle$ while U in Eq. (37) has $+\,|-\hat{x}\rangle$. They differ only by the phase of one basis ket, which is a free choice, so both are valid. For a change of basis with our convention for $|\pm\hat{x}\rangle$, always use U from Eq. (37).

Problem 24 (Rotation operator from Pauli matrices (PS3 Q1))**.** *Consider the rotation operator $R(\theta\hat{n}) = e^{-i\theta\vec{S}\cdot\hat{n}/\hbar}$, where θ is a general angle, $\hat{n}$ is a unit vector pointing in an arbitrary direction, and $\vec{S} = S_x\hat{x} + S_y\hat{y} + S_z\hat{z}$. In the S_z basis, $\vec{S} \to \frac{\hbar}{2}\vec{\sigma}$, with $\vec{\sigma} = \sigma_x\hat{x} + \sigma_y\hat{y} + \sigma_z\hat{z}$ and the Pauli matrices*

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

By considering the explicit Taylor series $e^x = \sum_j x^j/j!$, show that $R(\theta\hat{n})$ has the matrix representation (in the S_z basis)

$$R(\theta\hat{n}) \xrightarrow{S_z} \cos\frac{\theta}{2}\,\mathbb{1} - i\,\hat{n}\cdot\vec{\sigma}\,\sin\frac{\theta}{2}.$$

7 Angular Momentum

September 28, 2026

7.1 Non-commuting rotations and commutation relations

Definition 43 (Commutator)**.**

$$[A, B] = AB - BA. \tag{49}$$

A and B commute if $[A, B] = 0$, i.e. the order in which they act does not matter. Directly from the definition, $[B, A] = -[A, B]$ and $[A, BC] = B[A, C] + [A, B]C$.

Problem 25 (Commutator with a product on the right (PS3 Q5))**.** *Show that $[A, BC] = B[A, C] + [A, B]C$ for arbitrary operators A, B, C.*

Problem 26 (Commutator with a product on the left (PS3 Q6))**.** *Show that $[AB, C] = A[B, C] + [A, C]B$ for arbitrary operators A, B, C.*

Rotations about different axes do not commute: $R(\theta\hat{x})R(\theta\hat{y}) \neq R(\theta\hat{y})R(\theta\hat{x})$.

Remark 33. *Rotate a textbook by $90°$ about $\hat{x}$ then about $\hat{y}$ (right-hand rule). Repeat from the same start with $\hat{y}$ first. The spine ends up pointing in different directions.*

Remark 34 (Why the generators cannot commute)**.** *With $R_x = R(\theta\hat{x}) = e^{-i\theta J_x/\hbar}$ and $R_y = R(\theta\hat{y}) = e^{-i\theta J_y/\hbar}$, one might guess $R_xR_y = e^{-i\theta(J_x+J_y)/\hbar}$. That would force $R_xR_y = R_yR_x$, since $J_x + J_y = J_y + J_x$, which the textbook shows is false. The guess fails because*

$$e^A e^B = e^{A+B} \quad \text{if } [A, B] = 0, \tag{50}$$

but not in general otherwise. So the non-commutativity of rotations tells us $[J_x, J_y] = J_xJ_y - J_yJ_x \neq 0$.

Problem 27 (e^{A+B} versus e^Ae^B (PS3 Q9))**.** *Show that $e^{A+B} \neq e^Ae^B$ for two operators A and B, unless $[A, B] = 0$.*

Problem 28 ($[S_x, S_y]$ from Pauli matrices (PS3 Q2))**.** *Using the matrix representations of Problem 24, show that $[S_x, S_y] = i\hbar S_z$.*

Example ($[S_x, S_y]$ for spin-$\frac{1}{2}$)**.** *With $\hat{S}_j \xrightarrow{S_z} \frac{\hbar}{2}\sigma_j$,*

$$[S_x, S_y] \xrightarrow{S_z} \left(\frac{\hbar}{2}\right)^2 (\sigma_x\sigma_y - \sigma_y\sigma_x) = \left(\frac{\hbar}{2}\right)^2 \left[\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}\right] = i\hbar\frac{\hbar}{2}\sigma_z \xleftarrow{S_z} i\hbar S_z,$$

so $[S_x, S_y] = i\hbar S_z$.

Result 15 (Angular momentum commutation relations)**.** *The same relation holds for any angular momentum, and cyclic permutations ($x \to y \to z \to x$) give the other two:*

$$[J_x, J_y] = i\hbar J_z, \qquad [J_y, J_z] = i\hbar J_x, \qquad [J_z, J_x] = i\hbar J_y, \tag{51}$$

or compactly $\vec{J} \times \vec{J} = i\hbar \vec{J}$, with $\vec{J} = J_x\hat{x} + J_y\hat{y} + J_z\hat{z}$ and $\times$ the vector cross product (e.g. $\hat{x} \times \hat{y} = \hat{z}$).

Problem 29 (Rotation about a tilted axis (PS3 Q3))**.** *Use the construction of Problem 24 and the commutation relation of Problem 28 to prove the identity*

$$R^\dagger(\phi\hat{y})\, R(\theta\hat{n})\, R(\phi\hat{y}) = R(\theta\hat{z}),$$

where the unit vector lies in the x-z plane: $\hat{n} = \sin\phi\,\hat{x} + \cos\phi\,\hat{z}$. Give a geometrical argument that explains why this identity should be valid.

Problem 30 (Eigenstates of $R(\theta\hat{n})$ (PS3 Q4))**.** *Use the result of Problem 29 to find the eigenstates of $R(\theta\hat{n})$, written in terms of the $\{|\pm\hat{z}\rangle\}$ basis.*

7.2 Compatible observables

Definition 44 (Degenerate eigenvalue)**.** *An eigenvalue is degenerate if more than one independent eigenstate has it. Those eigenstates span a* subspace*: a smaller vector space inside the full one, closed under linear combinations.*

Theorem 3. *If $[A, B] = 0$, then A and B have a common eigenbasis.*

Proof. Let $A\,|a\rangle = a\,|a\rangle$. Since $AB = BA$,

$$A\big(B\,|a\rangle\big) = BA\,|a\rangle = a\big(B\,|a\rangle\big),$$

so $B\,|a\rangle$ is also an eigenstate of A with eigenvalue a.

Nondegenerate case. If $|a\rangle$ is the only state with eigenvalue a, then $B\,|a\rangle = b\,|a\rangle$, so $|a\rangle$ is an eigenstate of both.

Degenerate case. If n states $|a, i\rangle$ share eigenvalue a, then B maps their subspace into itself, so there it is an $n \times n$ matrix $B_{ij} = \langle a, i|\, B\,|a, j\rangle$. Diagonalizing it (finding the basis in which it is diagonal) gives eigenstates

$$|a, b_j\rangle = \sum_i |a, i\rangle\,\langle a, i|a, b_j\rangle\,, \qquad A\,|a, b_j\rangle = a\,|a, b_j\rangle\,, \qquad B\,|a, b_j\rangle = b_j\,|a, b_j\rangle\,,$$

which are still eigenstates of A because they are combinations of states with the same eigenvalue a. □

Remark 35 (Block-diagonal picture)**.** *Why B cannot mix different eigenvalues of A: if $B\,|a, i\rangle = \sum_{a',j} \langle a', j|B|a, i\rangle\,|a', j\rangle$ contained states with $a' \neq a$, then*

$$A\big(B\,|a, i\rangle\big) = \sum_{a',j} \langle a', j|B|a, i\rangle\; a'\,|a', j\rangle\,,$$

which equals $a\big(B\,|a,i\rangle\big)$ *only if* $\langle a',j|B|a,i\rangle \propto \delta_{aa'}$. *So in the basis* $\{|a_i,j\rangle\}$, B *is block diagonal, with one block per eigenvalue of* A*:*

$$B \xrightarrow{\{|a_i,j\rangle\}} \begin{pmatrix} B(a_1) & 0 & \cdots \\ 0 & B(a_2) & \\ \vdots & & \ddots \end{pmatrix}, \qquad B_{kl}(a_i) = \langle a_i,k|B|a_i,l\rangle\,.$$

Diagonalizing each block separately gives the simultaneous eigenstates, $|a,b_j\rangle = \sum_i \langle a,i|a,b_j\rangle\,|a,i\rangle$.

Definition 45 (Compatible observables)**.** *Observables that commute, and so can have definite values at the same time. Otherwise they are incompatible.*

Remark 36. $[J_x,J_y] = i\hbar J_z \neq 0$*, so the components of angular momentum are incompatible: they have no common eigenbasis (for spin-*$\frac{1}{2}$*, no common eigenstate at all). This is the operator version of Experiment 3: putting* $|+\hat{z}\rangle$ *into an SGx gives a 50/50 mix of* $|\pm\hat{x}\rangle$*, so no state gives* $S_z = +\hbar/2$ *and* $S_x = +\hbar/2$ *with certainty. Preparing an eigenstate of one observable leaves the outcomes of an incompatible one uncertain; for* $|+\hat{z}\rangle$ *and* S_x *the uncertainty is maximal,* $\Delta S_x = \hbar/2$.

7.3 Uncertainty relations

Result 16 (Generalized uncertainty relation)**.** *For observables* A *and* B*, in any state,*

$$\Delta A\,\Delta B \geq \frac{1}{2}\,|\langle[A,B]\rangle|\,. \tag{52}$$

The derivation is in Townsend, Eq. (3.74). The bound only says something when $[A,B] \neq 0$.

Problem 31 (Uncertainty relation for a constant commutator (PS3 Q8))**.** *For two Hermitian operators* A *and* B *having commutation relation* $[A,B] = i$*, show that the following uncertainty relation holds:*

$$\Delta A\,\Delta B \geq \frac{1}{2},$$

where $(\Delta A)^2 = \langle A^2\rangle - \langle A\rangle^2$ *and* $(\Delta B)^2 = \langle B^2\rangle - \langle B\rangle^2$.

Result 17. *For angular momentum,* $[J_x,J_y] = i\hbar J_z$ *gives*

$$\Delta J_x\,\Delta J_y \geq \frac{\hbar}{2}\,|\langle J_z\rangle|\,. \tag{53}$$

Example. *For* $|+\hat{z}\rangle$, $\langle S_z\rangle = \hbar/2$*, and both* S_x *and* S_y *give* $\pm\hbar/2$ *with probability* $\frac{1}{2}$*, so* $\Delta S_x = \Delta S_y = \hbar/2$*. Then* $\Delta S_x\,\Delta S_y = \hbar^2/4 = \frac{\hbar}{2}\langle S_z\rangle$*: the bound is saturated.*

Remark 37. *Position and momentum will also turn out not to commute,* $[x,p_x] = i\hbar$*, which gives the Heisenberg uncertainty relation*

$$\Delta x\,\Delta p_x \geq \frac{\hbar}{2}. \tag{54}$$

7.4 Total angular momentum

Definition 46 (Total angular momentum)**.** *The generator $\vec{J}$ of Section 6.2 is the total angular momentum, the sum of the orbital and spin angular momenta:*

$$\vec{J} = \vec{L} + \vec{S}. \tag{55}$$

Definition 47 (Square of the angular momentum)**.**

$$\vec{J}^2 = J_x^2 + J_y^2 + J_z^2. \tag{56}$$

Proposition 19. *$\vec{J}^2$ commutes with every component of $\vec{J}$; in particular $[\vec{J}^2, J_z] = 0$.*

Proof. Using $[A, BC] = B[A, C] + [A, B]C$:

$$\begin{aligned}
[J_z, J_x^2] &= J_x[J_z, J_x] + [J_z, J_x]J_x = i\hbar\,(J_xJ_y + J_yJ_x)\,,\\
[J_z, J_y^2] &= J_y[J_z, J_y] + [J_z, J_y]J_y = -i\hbar\,(J_yJ_x + J_xJ_y)\,,\\
[J_z, J_z^2] &= 0,
\end{aligned}$$

using $[J_z, J_y] = -i\hbar J_x$. The first two cancel, so $[J_z, \vec{J}^2] = 0$. The same steps work for x and y. □

Remark 38. *$\vec{J}^2$ and J_z are compatible, so they share an eigenbasis.*

Definition 48 (Angular momentum eigenstates)**.** *The simultaneous eigenstates of $\vec{J}^2$ and J_z are labelled $|j, m\rangle$, with $\langle j, m|j, m'\rangle = \delta_{mm'}$. By convention their eigenvalues are written*

$$\vec{J}^2\,|j, m\rangle = \hbar^2 j(j+1)\,|j, m\rangle\,, \qquad J_z\,|j, m\rangle = \hbar m\,|j, m\rangle\,. \tag{57}$$

For spin, $j = s$, matching the labels $|s, m\rangle$ and the magnitude $|\vec{S}| = \hbar\sqrt{s(s+1)}$ used earlier. The allowed values of j and m are derived in Section 7.5.

7.5 Ladder operators

Definition 49 (Ladder operators)**.**

$$J_\pm = J_x \pm iJ_y, \qquad J_x = \frac{1}{2}\,(J_+ + J_-)\,, \qquad J_y = \frac{1}{2i}\,(J_+ - J_-)\,. \tag{58}$$

They are not Hermitian: $J_+^\dagger = J_x - iJ_y = J_-$.

Proposition 20.

$$[J_z, J_\pm] = \pm\hbar J_\pm. \tag{59}$$

Proof. $[J_z, J_\pm] = [J_z, J_x] \pm i[J_z, J_y] = i\hbar J_y \pm i(-i\hbar J_x) = \pm\hbar\,(J_x \pm iJ_y)$. □

Proposition 21. *$J_\pm\,|j, m\rangle$ is an eigenstate of J_z with eigenvalue $\hbar(m \pm 1)$, so $J_\pm\,|j, m\rangle \propto |j, m \pm 1\rangle$.*

Proof. From $[J_z, J_-] = -\hbar J_-$, $J_z J_- = J_- J_z - \hbar J_- = J_-(J_z - \hbar)$, so

$$J_z\big(J_- |j,m\rangle\big) = J_-(J_z - \hbar)\,|j,m\rangle = \hbar(m-1)\big(J_- |j,m\rangle\big).$$

The same steps with $J_z J_+ = J_+(J_z + \hbar)$ give $\hbar(m+1)$. Since $\vec{J}^2$ commutes with J_x and J_y, it commutes with $J_\pm$, so j is unchanged. □

Proposition 22 (Action of the ladder operators)**.**

$$J_\pm |j,m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}\;|j, m\pm 1\rangle\,. \tag{60}$$

J_- is the lowering operator *and J_+ the* raising operator*.*

Proof. Since $J_+ = J_-^\dagger$, the squared norm of $J_- |j,m\rangle$ is $\langle j,m|J_+ J_-|j,m\rangle$. Expanding,

$$J_+ J_- = (J_x + iJ_y)(J_x - iJ_y) = J_x^2 + J_y^2 - i[J_x, J_y] = \vec{J}^2 - J_z^2 + \hbar J_z,$$

so with Eq. (57),

$$\langle j,m|J_+ J_-|j,m\rangle = \hbar^2\left[j(j+1) - m^2 + m\right] = \hbar^2\left[j(j+1) - m(m-1)\right].$$

Taking the coefficient real and positive (a phase convention) gives the J_- result. The same steps with $J_- J_+ = \vec{J}^2 - J_z^2 - \hbar J_z$ give the J_+ result. □

Result 18 (Allowed values of j and m)**.** *The ladder stops at both ends:*

$$J_+ |j,j\rangle = \hbar\sqrt{j(j+1) - j(j+1)}\;|j,j+1\rangle = 0, \qquad J_- |j,-j\rangle = \hbar\sqrt{j(j+1) - j(j+1)}\;|j,-j-1\rangle = 0.$$

Starting from $m = j$ and lowering, $J_- |j,j\rangle \propto |j,j-1\rangle$, $J_-^2 |j,j\rangle \propto |j,j-2\rangle$, $\ldots$, $J_-^{2j} |j,j\rangle \propto |j,-j\rangle$. Reaching $-j$ in whole steps requires $2j$ to be a whole number, so

$$j = 0, \tfrac{1}{2}, 1, \tfrac{3}{2}, \ldots, \qquad m = -j, -j+1, \ldots, j-1, j, \tag{61}$$

giving $2j+1$ states for each j, as in Eq. (**??**).

Remark 39. *The ladder cannot run past these ends because squared norms are non-negative: $j(j+1) - m(m\pm 1) \geq 0$ keeps $-j \leq m \leq j$.*

7.6 Matrix elements

Result 19. *In the $\{|j,m\rangle\}$ basis,*

$$\langle j,m'|J_\pm|j,m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}\,\delta_{m',m\pm 1}, \qquad \langle j,m'|J_z|j,m\rangle = \hbar m\,\delta_{m'm}. \tag{62}$$

Example (Spin 1)**.** *For $j = 1$ there are three states, $m = 1, 0, -1$ (e.g. the deuteron). Ordering the basis as $\{|1,1\rangle, |1,0\rangle, |1,-1\rangle\}$,*

$$J_z \longrightarrow \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad J_+ \longrightarrow \hbar \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix}, \qquad J_- \longrightarrow \hbar \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}.$$

With $J_x = \frac{1}{2}(J_+ + J_-)$ and $J_y = \frac{1}{2i}(J_+ - J_-)$,

$$J_x \longrightarrow \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad J_y \longrightarrow \frac{i\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Check $[J_x, J_y] = i\hbar J_z$:

$$[J_x, J_y] \longrightarrow \frac{i\hbar^2}{2}\left[\begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix} - \begin{pmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}\right] = i\hbar\cdot\hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \longleftarrow i\hbar J_z.$$

Problem 32 (Spin-1 matrices in ascending order (PS3 Q7))**.** *Construct matrix representations for each of the three spin-1 operators J_x, J_y, J_z in the basis of states $\{|1,-1\rangle, |1,0\rangle, |1,+1\rangle\}$, where the state $|j,m\rangle$ is a simultaneous eigenstate of $\vec{J}^2 = \vec{J}\cdot\vec{J} = J_x^2 + J_y^2 + J_z^2$ with eigenvalue $\hbar^2 j(j+1) = 2\hbar^2$ and of J_z, with associated eigenvalue $\hbar m$.*

Problem 33 (Anisotropy operator for $j = \frac{3}{2}$ (PS3 Q10))**.** *Find the matrix representation of the operator $A = J_+^2 - J_-^2$, where $J_\pm = J_x \pm iJ_y$, in the J_z eigenbasis for $j = \frac{3}{2}$. This operator arises in the energy of certain magnetic materials due to anisotropy (broken rotational symmetry about the z-axis).*

7.7 Rotations by 2π: fermions and bosons

Using the spectral form (Section 6.4) with j fixed,

$$R(2\pi\hat{z}) = e^{-2\pi i J_z/\hbar} \sum_m |j,m\rangle\langle j,m| = \sum_m e^{-2\pi i m} |j,m\rangle\langle j,m|.$$

Result 20. *$R(2\pi\hat{z}) = \mathbb{1}$ only if every m is an integer, i.e. $j = 0, 1, 2, \ldots$. For half-integer j, every m is a half-integer and $R(2\pi\hat{z}) = -\mathbb{1}$.*

Remark 40. *If we demanded that a 2π rotation leave every state unchanged, j would have to be an integer. But electrons have $j = \frac{1}{2}$, with $m = \pm\frac{1}{2}$ and $S_z = \pm\hbar/2$, so $R(2\pi\hat{z}) = e^{\mp i\pi} = -\mathbb{1}$ (Remark 28). For a single spin-$\frac{1}{2}$ particle this is a global phase, $R(2\pi\hat{z})|\psi\rangle = -|\psi\rangle = e^{i\pi}|\psi\rangle$, with no effect on observables. For several particles, this sign for half-integer spin results in the Pauli exclusion principle.*

Definition 50 (Fermions and bosons)**.** *Particles with half-integer spin $j = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$ are* fermions *and obey Pauli exclusion. Particles with integer spin $j = 0, 1, 2, \ldots$ are* bosons.

Remark 41. *Loosely, this connection to rotations is related to the* exchange *of two identical particles: swapping them is like rotating one around the other.*

7.8 Atomic quantum numbers

Definition 51 (Atomic quantum numbers)**.** *An electron in an atom is labelled* $|n, l, m_l, s, m_s\rangle$*, where*

- $n = 1, 2, 3, \ldots$ *is the* principal *quantum number: the size and energy of the electron cloud.*
- $l = 0, 1, \ldots, n-1$ *is the* azimuthal *quantum number: the shape of the electron cloud, with* $|\vec{L}| = \hbar\sqrt{l(l+1)}$*. The values* $l = 0, 1, 2, 3$ *are called s, p, d, f.*
- $m_l = -l, -l+1, \ldots, +l$ *is the* magnetic *quantum number: the spatial orientation of the electron cloud, with* $L_z = m_l\hbar$*.*
- $s = \frac{1}{2}$ *is the* spin *quantum number: the intrinsic angular momentum of the electron, with* $m_s = \pm\frac{1}{2}$ *giving* $S_z = m_s\hbar$*.*

The total angular momentum $\vec{J} = \vec{L} + \vec{S}$ *has its own quantum number* j*, running from* $|l - s|$ *to* $l + s$ *in steps of 1. For an electron,* $j = l \pm \frac{1}{2}$ *(only* $j = \frac{1}{2}$ *when* $l = 0$*).*

Remark 42. *No two electrons in an atom share all of n, l,* m_l*,* m_s *(Pauli exclusion principle). A subshell l has* $2l + 1$ *values of* m_l*, each with two values of* m_s*, so it holds* $2(2l + 1)$ *electrons: 2 for s, 6 for p, 10 for d, 14 for f.*

Remark 43. *Two notation clashes: the subshell name s (*$l = 0$*) is not the spin quantum number s, and the quantum number j is not the outcome label j in* $|j\rangle$*.*

8 Time Evolution

September 30 - October 16, 2026

Chapter 4

8.1 The time-evolution operator and the Hamiltonian

A small rotation is generated by total angular momentum (Section 6.2),

$$R(d\phi\,\hat{n}) = \mathbb{1} - \frac{i}{\hbar}\left(\vec{J}\cdot\hat{n}\right)d\phi, \qquad R(\phi\hat{n})\,|\psi_0\rangle = |\psi_\phi\rangle\,,$$

where ϕ is the parameter that labels how far the state has been rotated. Time evolution works the same way, with t as the parameter. Now it's time to move in time, since so far we are measuring states in the current moment, but in reality, particles change through time! So for that we need to introduce a new operator.

Definition 52 (Time-evolution operator)**.** *The linear operator* $U(t)$ *that takes a state from time* 0 *to time t:*

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,. \tag{63}$$

Remark 44. *Not to be confused with the change-of-basis matrix U of Section 5.4. Both are unitary, which is where the shared letter comes from.*

Proposition 23. *$U(t)$ is unitary: $U^\dagger(t)U(t) = \mathbb{1}$.*

Proof. Probabilities must still add to 1 at every time,

$$\langle\psi(t)|\psi(t)\rangle = \langle\psi(0)|\, U^\dagger(t)U(t)\,|\psi(0)\rangle = \langle\psi(0)|\psi(0)\rangle = 1$$

for every $|\psi(0)\rangle$, which forces $U^\dagger U = \mathbb{1}$. □

Remark 45. *$U(t)$ is not derived from anything: it is assumed to be linear, to conserve probability (so unitary, as shown above), to satisfy $U(0) = \mathbb{1}$, and to compose as $U(t_1 + t_2) = U(t_1)U(t_2)$. These conditions lead us to its Taylor expansion, and from that its actual formula.*

By composition, evolving for t is N small steps of t/N:

$$U(t) = \underbrace{U\left(\tfrac{t}{N}\right) U\left(\tfrac{t}{N}\right) \cdots U\left(\tfrac{t}{N}\right)}_{N \text{ factors}} = \left[U\left(\tfrac{t}{N}\right)\right]^N, \qquad U\left(\tfrac{t}{N}\right) = \mathbb{1} + U'(0)\,\frac{t}{N} + \mathcal{O}\left(\frac{1}{N^2}\right).$$

Definition 53 (Hamiltonian)**.** *The Hamiltonian*[3]*,*

$$H \equiv i\hbar\, U'(0). \tag{64}$$

The i makes H Hermitian (Problem 36), and the $\hbar$ gives it units $[H] = [\hbar/t] =$ energy, so H is identified as the energy operator.

Substituting $U'(0) = -\frac{i}{\hbar}H$,

$$U\left(\tfrac{t}{N}\right) = \mathbb{1} - \frac{i}{\hbar}H\,\frac{t}{N} + \mathcal{O}\left(\frac{1}{N^2}\right). \tag{65}$$

Problem 34 (Generator of translations (PS4 Q1))**.** *Show that, if a state $|x\rangle$ is labelled by a parameter x, then the operator describing translations in x, satisfying $U(\Delta x)\,|x\rangle = |x + \Delta x\rangle$, is given by $U(\Delta x) = e^{-i\Delta x\, p/\hbar}$, where p is the generator of translations in x.*

Problem 35 (Units of the generator of translations (PS4 Q2))**.** *Show that if x in Problem 34 has dimensions of length, then p has dimensions of momentum.*

Remark 46. *The same argument as for J_z (Section 6.2) shows that unitarity of U forces $H = H^\dagger$, so energy is an observable (Principle 2).*

Problem 36 (Hermiticity of H (PS4 Q3))**.** *Show that $H = H^\dagger$ follows from the unitarity of the time-evolution operator $U(t) = e^{-iHt/\hbar}$.*

Problem 37 (Eigenvalues of H (PS4 Q4))**.** *Based on the result of Problem 36 and the definition of $U(t)$, show that the eigenvalues of H must be real numbers and that they have units of energy.*

[3] Also referred to as the generator of time evolution.

Proposition 24 (Schrödinger equation)**.**

$$\frac{d}{dt}\left|\psi(t)\right\rangle = -\frac{i}{\hbar}H\left|\psi(t)\right\rangle. \tag{66}$$

Proof. By composition, $U(t+\Delta t) = U(\Delta t)\,U(t)$. The small step $U(\Delta t)$ is Eq. (65) with $t/N \to \Delta t$, so

$$U(t+\Delta t) = \left(\mathbb{1} - \frac{i}{\hbar}H\Delta t + \mathcal{O}(\Delta t^2)\right)U(t) = U(t) - \frac{i}{\hbar}H\Delta t\,U(t) + \mathcal{O}(\Delta t^2).$$

Using the limit definition of a derivative[4],

$$\frac{dU}{dt} = \lim_{\Delta t\to 0}\frac{U(t+\Delta t)-U(t)}{\Delta t} = -\frac{i}{\hbar}HU(t).$$

Acting on $\left|\psi(0)\right\rangle$, which does not depend on t, and using $\left|\psi(t)\right\rangle = U(t)\left|\psi(0)\right\rangle$,

$$\frac{d}{dt}\left|\psi(t)\right\rangle = \frac{dU}{dt}\left|\psi(0)\right\rangle = -\frac{i}{\hbar}H\,U(t)\left|\psi(0)\right\rangle = -\frac{i}{\hbar}H\left|\psi(t)\right\rangle.$$

□

8.1.1 When H is constant

For constant H, from (65),

$$U(t) = \lim_{N\to\infty}\left(\mathbb{1} - \frac{i}{\hbar}H\,\frac{t}{N}\right)^N = e^{-iHt/\hbar}.$$

Result 21 (Time-independent Hamiltonian)**.** *If H does not depend on time,*

$$U(t) = e^{-iHt/\hbar}. \tag{67}$$

8.1.2 When H depends on time

If $H = H(t)$, then $e^{-iH(t)t/\hbar}$ does not solve the Schrödinger equation in general.

Problem 38 (Time-dependent Hamiltonian (PS4 Q6))**.** *Show that, for an explicitly time-dependent Hamiltonian $H(t)$, and provided $[H(t), H(t')] = 0$, the time-evolution operator can be written as*

$$U(t) = \exp\left\{-\frac{i}{\hbar}\int_0^t dt'\,H(t')\right\}. \tag{68}$$

Remark 47. *Physically, $[H(t), H(t')] = 0$ means the eigenstates of H never change, only the energies.*

Example. *Let a particle be placed in $\vec{B} = B_0(t)\hat{z}$: its spin always rotates about $\hat{z}$, so the small rotations add into one rotation by $\int_0^t \omega_0(t')\,dt'$. If for example the field changes direction after some time, the eigenstates of H change $\therefore [H(t), H(t')] \neq 0$, so the spin of the particle rotates about a different axis. Then the order matters, and no single exponential works.*

[4] The $\mathcal{O}(\Delta t^2)$ term goes to zero faster than Δt so it vanishes.

8.2 Stationary states

Definition 54 (Stationary state)**.** *A state that does not evolve into a physically different state in time. These are the eigenstates of time-independent H, the energy eigenstates $H|E\rangle = E|E\rangle$.*

Proposition 25.

$$U(t)|E\rangle = e^{-iEt/\hbar}|E\rangle\,. \tag{69}$$

This is a global phase, so $|E\rangle$ and $U(t)|E\rangle$ are the same physical state for all t.

Proof. Proposition 14 with $f(x) = e^{-ixt/\hbar}$. □

Result 22 (Evolution of a general state)**.** *H is Hermitian, so its eigenstates $|E_n\rangle$, where n labels the different energies (e.g. for $H = \omega_0 S_z$, $|E_\pm\rangle = |\pm\hat{z}\rangle$), form an orthonormal basis, and the completeness relation (Section 4) reads $\mathbb{1} = \sum_n |E_n\rangle\langle E_n|$. Inserting it,*

$$|\psi(0)\rangle = \mathbb{1}|\psi(0)\rangle = \sum_n |E_n\rangle\langle E_n|\psi(0)\rangle = \sum_n \langle E_n|\psi(0)\rangle |E_n\rangle\,.$$

$U(t)$ is linear, so it acts on each term separately, and each $|E_n\rangle$ only picks up its phase:

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle = \sum_n \langle E_n|\psi(0)\rangle\ U(t)|E_n\rangle = \sum_n \langle E_n|\psi(0)\rangle\ e^{-iE_nt/\hbar}|E_n\rangle\,. \tag{70}$$

8.3 Time-dependent expectation values

Proposition 26 (Time derivative of an expectation value)**.**

$$\frac{d}{dt}\langle A\rangle = \frac{i}{\hbar}\langle[H,A]\rangle + \langle\frac{\partial A}{\partial t}\rangle\,. \tag{71}$$

This tells us how the average of any observable changes in time.[5]

Proof. $\langle A\rangle = \langle\psi(t)|A|\psi(t)\rangle$ is a product of three things that can depend on t: the bra, A and the ket. The product rule gives one term for each:

$$\frac{d}{dt}\langle A\rangle = \left(\frac{d}{dt}\langle\psi(t)|\right)A|\psi(t)\rangle + \langle\psi(t)|A\left(\frac{d}{dt}|\psi(t)\rangle\right) + \langle\psi(t)|\frac{\partial A}{\partial t}|\psi(t)\rangle\,.$$

The Schrödinger equation and its bra (using $H^\dagger = H$) give $\frac{d}{dt}|\psi\rangle = -\frac{i}{\hbar}H|\psi\rangle$ and $\frac{d}{dt}\langle\psi| = \frac{i}{\hbar}\langle\psi|H$, so the first two terms are $\frac{i}{\hbar}\langle\psi|(HA - AH)|\psi\rangle$. □

Remark 48. *The $\partial A/\partial t$ term is only there if the operator itself has explicit time dependence, e.g. a time-dependent field.*

[5] Also called the *equation of motion* for $\langle A\rangle$, since it plays the role Newton's second law plays classically.

Definition 55 (Constant of the motion)**.** *An observable A with $\partial A/\partial t = 0$ and $[H, A] = 0$. Then $\frac{d}{dt}\langle A\rangle = 0$, so A is a conserved quantity.*

Example. *Take $A = H$. Then $[H, A] = [H, H] = 0$ always, so whenever $\partial H/\partial t = 0$, energy is a constant of the motion.*

Problem 39 (Conservation of energy (PS4 Q5))**.** *Show that $[H, U(t)] = 0$, and hence that the average energy (expectation value of H) is conserved, by proving that $E = \langle H\rangle = \langle\psi(t)|\, H\, |\psi(t)\rangle = \langle\psi(0)|\, H\, |\psi(0)\rangle$ for any time t.*

8.4 Spin in a static field: precession

Definition 56 (Precession)**.** *The phenomenon where the axis of a spinning object rotates about another axis due to external torques.*

8.4.1 Time evolution as a rotation

Put an electron ($q = -|e|$) in a uniform field $\vec{B} = B_0\hat{n}$ along any axis $\hat{n}$. With $\vec{\mu} = -g\mu_B\vec{S}/\hbar$ and $S_n = \vec{S}\cdot\hat{n}$,

$$H_0 = -\vec{\mu}\cdot\vec{B} = \omega_0 S_n, \qquad \omega_0 = \frac{g\mu_B B_0}{\hbar}, \qquad [\omega_0] = \frac{[\text{energy}]}{[\hbar]} = [\text{frequency}]. \tag{72}$$

Since $[H_0, S_n] = 0$, S_n is a constant of the motion. The energy eigenstates are

$$H_0\,|\pm\hat{n}\rangle = \pm\frac{\hbar\omega_0}{2}\,|\pm\hat{n}\rangle\,.$$

Result 23 (Time evolution is a rotation)**.**

$$U(t) = e^{-i\omega_0 t S_n/\hbar} = R(\phi\hat{n})\Big|_{\phi=\omega_0 t}, \qquad U(t)\,|\pm\hat{n}\rangle = e^{\mp i\omega_0 t/2}\,|\pm\hat{n}\rangle\,. \tag{73}$$

Evolving in a field along $\hat{n}$ is the same as rotating about $\hat{n}$ by $\phi = \omega_0 t$. The states $|\pm\hat{n}\rangle$ are stationary, equivalently invariant under rotations about $\hat{n}$. The examples below take $\hat{n} = \hat{z}$.

8.4.2 Probabilities

Example (Starting in $|+\hat{x}\rangle$)**.** *The spin starts along $+\hat{x}$ in a field along $\hat{z}$. Find the probability that it still points along $+\hat{x}$ at time t.*

Expand in energy eigenstates. *The energy eigenstates are $|\pm\hat{z}\rangle$, with $E_\pm = \pm\hbar\omega_0/2$, and*

$$|\psi(0)\rangle = |+\hat{x}\rangle = \frac{1}{\sqrt{2}}\,|+\hat{z}\rangle + \frac{1}{\sqrt{2}}\,|-\hat{z}\rangle\,.$$

Attach the phases $e^{-iE_\pm t/\hbar} = e^{\mp i\omega_0 t/2}$, *as in Eq. (70):*

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(e^{-i\omega_0 t/2}\,|+\hat{z}\rangle + e^{i\omega_0 t/2}\,|-\hat{z}\rangle\right).$$

Born rule, *with* $\langle +\hat{x}| \pm \hat{z}\rangle = \frac{1}{\sqrt{2}}$*:*

$$P_{+\hat{x}}(t) = |\langle +\hat{x}|\psi(t)\rangle|^2 = \left|\frac{1}{2}e^{-i\omega_0 t/2} + \frac{1}{2}e^{i\omega_0 t/2}\right|^2 = \cos^2\frac{\omega_0 t}{2} = \frac{1}{2}\left(1+\cos\omega_0 t\right). \tag{74}$$

Problem 40 (Precession about $\hat{y}$ (PS4 Q7))**.** *For a spin-*$\frac{1}{2}$ *placed in a magnetic field* $\vec{B} = B_0\hat{y}$*, and initially oriented along the* $\hat{x}$ *direction, find the probability to measure the spin oriented along* $+\hat{z}$ *at a later time* t*.*

8.4.3 Average spin

Two ways to get $\langle S_x(t)\rangle$ and $\langle S_y(t)\rangle$ for the spin starting in $|+\hat{x}\rangle$ (field along $\hat{z}$): from the state, or from Proposition 26 without ever computing the state.

Example (Method 1: from the state)**.** *Use* $\langle A\rangle = \psi^\dagger A\,\psi$ *(row times matrix times column, from Matrix Mechanics), with* $S_x \xrightarrow{S_z} \frac{\hbar}{2}\sigma_x$ *and the state from the previous example,* $\psi(t) \xrightarrow{S_z} \frac{1}{\sqrt{2}}\left(e^{-i\omega_0 t/2}, e^{i\omega_0 t/2}\right)^T$*:*

$$\langle S_x(t)\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}e^{i\omega_0 t/2} & e^{-i\omega_0 t/2}\end{pmatrix}\frac{\hbar}{2}\begin{pmatrix}0&1\\1&0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}e^{-i\omega_0 t/2}\\ e^{i\omega_0 t/2}\end{pmatrix}$$

$$= \frac{\hbar}{4}\left(e^{i\omega_0 t} + e^{-i\omega_0 t}\right) = \frac{\hbar}{2}\cos\omega_0 t.$$

Alternatively, from the probabilities,

$$\langle S_x(t)\rangle = \frac{\hbar}{2}\left(P_{+\hat{x}} - P_{-\hat{x}}\right) = \frac{\hbar}{2}\left(2P_{+\hat{x}} - 1\right) = \frac{\hbar}{2}\cos\omega_0 t.$$

The same matrix product with $S_y \xrightarrow{S_z} \frac{\hbar}{2}\sigma_y$ *gives* $\langle S_y(t)\rangle = \frac{\hbar}{2}\sin\omega_0 t$*.*

Example (Method 2: from $d\langle A\rangle/dt$)**.** *With* $H_0 = \omega_0 S_z$*,* $[S_z, S_x] = i\hbar S_y$ *and* $[S_z, S_y] = -i\hbar S_x$*:*

$$\frac{d}{dt}\langle S_x\rangle = \frac{i}{\hbar}\langle[\omega_0 S_z, S_x]\rangle = -\omega_0\langle S_y\rangle, \tag{75}$$

$$\frac{d}{dt}\langle S_y\rangle = \frac{i}{\hbar}\langle[\omega_0 S_z, S_y]\rangle = \omega_0\langle S_x\rangle. \tag{76}$$

Differentiating Eq. (75) and using Eq. (76),

$$\frac{d^2}{dt^2}\langle S_x\rangle = -\omega_0\frac{d}{dt}\langle S_y\rangle = -\omega_0^2\langle S_x\rangle,$$

with general solution $\langle S_x\rangle = A\cos\omega_0 t + B\sin\omega_0 t$*. The initial conditions are* $\langle S_x(0)\rangle = \hbar/2$ *and, from Eq. (75),* $\frac{d}{dt}\langle S_x\rangle\big|_{t=0} = -\omega_0\langle S_y(0)\rangle = 0$*, so* $A = \hbar/2$*,* $B = 0$*. Then* $\langle S_y\rangle = -\frac{1}{\omega_0}\frac{d}{dt}\langle S_x\rangle$ *gives*

$$\langle S_x(t)\rangle = \frac{\hbar}{2}\cos\omega_0 t, \qquad \langle S_y(t)\rangle = \frac{\hbar}{2}\sin\omega_0 t.$$

Result 24 (Precession)**.**

$$\langle \vec{S}(t) \rangle = \frac{\hbar}{2} \left(\cos \omega_0 t \, \hat{x} + \sin \omega_0 t \, \hat{y} \right), \qquad \langle S_z \rangle = 0. \tag{77}$$

The average spin rotates counterclockwise about $\hat{z}$ at angular frequency ω_0 (Larmor precession), while $\langle S_z \rangle$ stays constant as a constant of the motion should. Physically, the field exerts a torque $\vec{\mu} \times \vec{B}$ perpendicular to the spin, so instead of lining up with $\vec{B}$ the spin circles around it, like a tilted spinning top circling around gravity. It completes one turn every $2\pi/\omega_0$.

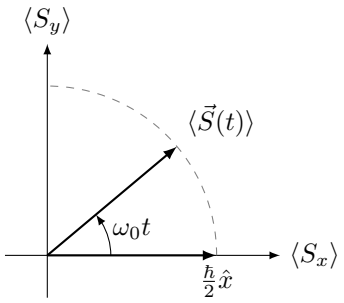


Figure 3: Precession of $\langle \vec{S} \rangle$ in the x-y plane, starting from $|+\hat{x}\rangle$.

Problem 41 ($\langle S_x(t) \rangle$ for precession about $\hat{y}$ (PS4 Q8))**.** *For the same initial conditions and magnetic field considered in Problem 40, find the expectation value $\langle S_x(t) \rangle$.*

Problem 42 (Bloch vector (PS4 Q9))**.** *Consider a spin-$\frac{1}{2}$ precessing in a magnetic field oriented along the z-direction. Define Pauli operators X, Y, Z in terms of the spin-$\frac{1}{2}$ operators by $S_x = \frac{\hbar}{2}X$, $S_y = \frac{\hbar}{2}Y$, $S_z = \frac{\hbar}{2}Z$. Find the time dependence of the components of the Bloch vector $\vec{s}(t) = (\langle X(t) \rangle, \langle Y(t) \rangle, \langle Z(t) \rangle)^T$ with the initial condition $\vec{s}(0) = (\cos\theta, 0, \sin\theta)^T$ by deriving and then solving the coupled equations of motion for the components.*

8.5 Magnetic resonance

Definition 57 (Magnetic resonance)**.** *Driving transitions between spin states with an oscillating field tuned to the precession frequency. Applications:*

- Nuclear magnetic resonance *(NMR), on nuclear spin $\vec{I}$: e.g. medical imaging (MRI), which maps local water content through resonant absorption of radio-frequency ($\sim$MHz) radiation.*[6]

[6] MRI is NMR used for imaging. It was renamed MRI because patients were scared of the word "nuclear".

- Electron spin resonance *(ESR), on electron spin $\vec{S}$: e.g. characterizing materials with microwave ($\sim$GHz) radiation.*
- Qubit control*: single-qubit gates (unitary operations) on qubits made from electron spins, nuclear spins, ions, neutral atoms or superconducting circuits.*

The field is a strong longitudinal part plus a weaker oscillating transverse part,

$$\vec{B}(t) = B_0\hat{z} + B_1 \cos\omega t\, \hat{x}, \qquad B_0 \gg B_1, \tag{78}$$

so the Hamiltonian now has explicit time dependence:

$$H(t) = -\vec{\mu}\cdot\vec{B}(t) = \omega_0 S_z + \Omega\cos(\omega t)\, S_x, \tag{79}$$

with, using the magnetons defined with the Stern-Gerlach experiment,

$$\begin{aligned} \text{ESR } (q=-|e|): &\quad \omega_0 = \frac{g\mu_B B_0}{\hbar}, \qquad \Omega = \frac{g\mu_B B_1}{\hbar}, \\ \text{NMR } (q=+|e|): &\quad \omega_0 = -\frac{g_I\mu_N B_0}{\hbar}, \qquad \Omega = -\frac{g_I\mu_N B_1}{\hbar}. \end{aligned}$$

Remark 49. *For $B_0 \sim 1$ T, $\omega_0/2\pi$ is in the GHz range for ESR ($g \approx 2$) and the MHz range for NMR, since $\mu_N \approx \mu_B/1836$.*

Remark 50 (Angular vs linear frequency)**.** *ω_0 is an angular frequency, in rad/s. $\omega_0/2\pi$ is the linear frequency, in Hz = cycles/s, with 1 cycle = 2π rad.*

Remark 51. *With H time-dependent, $U(t) = e^{-iHt/\hbar}$ no longer holds, and we must work with the Schrödinger equation directly, $\partial_t|\psi(t)\rangle = -\frac{i}{\hbar}H(t)|\psi(t)\rangle$.*

8.5.1 Going into the rotating frame

Split the Hamiltonian into a large static part and a small driving part,

$$H(t) = H_0 + V(t), \qquad H_0 = \omega_0 S_z, \qquad V(t) = \Omega\cos(\omega t)\, S_x.$$

Since $\omega_0 \gg \Omega$, most of the evolution is the fast precession from H_0. Factor it out:

$$|\psi(t)\rangle = U_0(t)\,|\tilde{\psi}(t)\rangle\,, \qquad U_0(t) = e^{-iH_0 t/\hbar}, \tag{80}$$

where U_0 is the "fast" part and $|\tilde{\psi}(t)\rangle$ the "slow" part.

Proposition 27 (Schrödinger equation in the rotating frame)**.**

$$\partial_t|\tilde{\psi}(t)\rangle = -\frac{i}{\hbar}\tilde{V}(t)\,|\tilde{\psi}(t)\rangle\,, \qquad \tilde{V}(t) = U_0^\dagger(t)V(t)U_0(t). \tag{81}$$

Proof. Substituting into the Schrödinger equation with $\dot{U}_0 = -\frac{i}{\hbar}H_0U_0$,

$$-\frac{i}{\hbar}H_0U_0\,|\tilde{\psi}\rangle + U_0\,\partial_t|\tilde{\psi}\rangle = -\frac{i}{\hbar}\,(H_0+V)\,U_0\,|\tilde{\psi}\rangle\,.$$

The H_0 terms cancel. Multiplying on the left by $U_0^\dagger$ gives the result. □

Definition 58 (Rotating frame)**.** *Since $U_0(t) = e^{-i\omega_0 t S_z/\hbar} = R(\omega_0 t\,\hat{z})$ is a rotation about $\hat{z}$ by the time-dependent angle $\omega_0 t$, describing the system by $|\tilde{\psi}(t)\rangle$ is called going into the rotating frame.*

To compute $\tilde{V}$, write $S_x = \frac{1}{2}(S_+ + S_-)$ with the spin-$\frac{1}{2}$ ladder operators (Section 7.5),

$$S_+ = \hbar\,|+\hat{z}\rangle\langle -\hat{z}|\,, \qquad S_- = \hbar\,|-\hat{z}\rangle\langle +\hat{z}|\,.$$

Lemma 4.

$$e^{i\omega_0 t S_z/\hbar}\, S_\pm\, e^{-i\omega_0 t S_z/\hbar} = e^{\pm i\omega_0 t} S_\pm. \tag{82}$$

Proof. For S_+: $e^{-i\omega_0 t S_z/\hbar}$ acting right on $|-\hat{z}\rangle$ gives $e^{i\omega_0 t/2}$, and $e^{i\omega_0 t S_z/\hbar}$ acting left on $|+\hat{z}\rangle$ gives $e^{i\omega_0 t/2}$, a total of $e^{i\omega_0 t}$. Swapping the roles gives $e^{-i\omega_0 t}$ for S_-. □

With $\cos\omega t = \frac{1}{2}\left(e^{i\omega t} + e^{-i\omega t}\right)$,

$$\begin{aligned}\tilde{V}(t) &= \frac{\Omega}{4}\left(e^{i\omega t} + e^{-i\omega t}\right)\left(e^{i\omega_0 t} S_+ + e^{-i\omega_0 t} S_-\right) \\ &= \frac{\Omega}{4}\Big(\underbrace{e^{-i(\omega-\omega_0)t} S_+ + e^{i(\omega-\omega_0)t} S_-}_{\text{slow for } \omega\approx\omega_0} + \underbrace{e^{i(\omega+\omega_0)t} S_+ + e^{-i(\omega+\omega_0)t} S_-}_{\text{fast}}\Big).\end{aligned} \tag{83}$$

8.5.2 Rotating-wave approximation

Integrating the rotating-frame equation,

$$|\tilde{\psi}(t)\rangle - |\tilde{\psi}(0)\rangle = -\frac{i}{\hbar}\int_0^t dt'\,\tilde{V}(t')\,|\tilde{\psi}(t')\rangle\,.$$

The fast terms $\sim e^{\pm i(\omega+\omega_0)t}$ oscillate many times while $|\tilde{\psi}\rangle$ barely changes, so they average to zero on a time scale $t \gtrsim 1/(\omega+\omega_0)$. Near resonance ($\omega \approx \omega_0$) the slow terms $\sim e^{\pm i(\omega-\omega_0)t}$ dominate.

Definition 59 (Rotating-wave approximation (RWA))**.** *Dropping the terms $\sim e^{\pm i(\omega+\omega_0)t}$. The corrections are small when $\Omega/\omega_0 \ll 1$, i.e. $B_0 \gg B_1$. Then*

$$\tilde{V}(t) \simeq \frac{\Omega}{4}\left(e^{-i\delta t} S_+ + e^{i\delta t} S_-\right). \tag{84}$$

Definition 60 (Detuning)**.** *$\delta = \omega - \omega_0$, how far the drive is from resonance.*

Result 25 (On resonance)**.** *For $\omega = \omega_0$, $\delta = 0$ and $\tilde{V}$ is time-independent:*

$$\tilde{V} \simeq \frac{\Omega}{4}(S_+ + S_-) = \frac{\Omega}{2} S_x, \qquad |\tilde{\psi}(t)\rangle \simeq e^{-i\Omega t S_x/2\hbar}\,|\psi(0)\rangle = R\left(\tfrac{\Omega t}{2}\hat{x}\right)|\psi(0)\rangle\,, \tag{85}$$

using $|\tilde{\psi}(0)\rangle = |\psi(0)\rangle$ since $U_0(0) = \mathbb{1}$. In the rotating frame the spin simply rotates about $\hat{x}$ at angular frequency $\Omega/2$.

Problem 43 (Three-level atom in a Λ configuration (PS5 Q1))**.** *A three-level atom has one high-energy state $|1\rangle$ and two lower-energy states $|0\rangle$, $|2\rangle$. It is driven on resonance by external time-dependent fields, giving*

$$H(t) = H_0 + V(t), \qquad H_0 = \sum_{j\in\{0,1,2\}} \hbar\omega_j \left|j\right\rangle\left\langle j\right|,$$

$$V(t) = \hbar\Omega\cos(\omega_{10}t)\left(\left|1\right\rangle\left\langle 0\right| + \left|0\right\rangle\left\langle 1\right|\right) + \hbar\Omega\cos(\omega_{21}t+\phi)\left(\left|1\right\rangle\left\langle 2\right| + \left|2\right\rangle\left\langle 1\right|\right),$$

with $\omega_{ij} = \omega_i - \omega_j$, Ω the drive amplitude and ϕ a phase shift. Assume $|\omega_{01}| \gg |\Omega|$ and $|\omega_{21}| \gg |\Omega|$, so oscillations at frequencies $\sim 2\omega_{01}$, $\sim 2\omega_{12}$ can be averaged over.

(a) Starting from $\partial_t \left|\psi(t)\right\rangle = -\frac{i}{\hbar}H(t)\left|\psi(t)\right\rangle$, derive $\partial_t \left|\tilde{\psi}(t)\right\rangle = -\frac{i}{\hbar}\tilde{V}(t)\left|\tilde{\psi}(t)\right\rangle$, where $\left|\psi(t)\right\rangle = U_0(t)\left|\tilde{\psi}(t)\right\rangle$ and $U_0(t) = e^{-iH_0t/\hbar}$. Give $\tilde{V}(t)$ in terms of $V(t)$ and $U_0(t)$.

(b) Show that within the rotating-wave approximation,

$$\tilde{V}(t) \simeq V_{\text{RWA}} = \frac{\hbar\Omega}{2}\left(\left|1\right\rangle\left\langle 0\right| + \left|0\right\rangle\left\langle 1\right| + e^{i\phi}\left|1\right\rangle\left\langle 2\right| + e^{-i\phi}\left|2\right\rangle\left\langle 1\right|\right).$$

(c) Show that the dark state $\left|D\right\rangle = \frac{1}{\sqrt{2}}\left(\left|0\right\rangle - e^{-i\phi}\left|2\right\rangle\right)$ is an eigenstate of V_{RWA}.

(d) With $\left|+\right\rangle = \frac{1}{\sqrt{2}}\left(\left|0\right\rangle + e^{-i\phi}\left|2\right\rangle\right)$, show that $V_{\text{RWA}} = \frac{\hbar\Omega}{\sqrt{2}}\left(\left|1\right\rangle\left\langle +\right| + \left|+\right\rangle\left\langle 1\right|\right)$.

(e) Show that $V_{\text{RWA}}\left|B_\pm\right\rangle = \pm\frac{\hbar\Omega}{\sqrt{2}}\left|B_\pm\right\rangle$ for the bright states $\left|B_\pm\right\rangle = \frac{1}{\sqrt{2}}\left|1\right\rangle \pm \frac{1}{2}\left(\left|0\right\rangle + e^{-i\phi}\left|2\right\rangle\right)$.

8.5.3 Rabi oscillations

Start in $\left|\psi(0)\right\rangle = \left|+\hat{z}\right\rangle = \frac{1}{\sqrt{2}}\left(\left|+\hat{x}\right\rangle + \left|-\hat{x}\right\rangle\right)$, i.e. expand in the eigenstates of $\tilde{V} \propto S_x$ as in Eq. (70). With $S_x\left|\pm\hat{x}\right\rangle = \pm\frac{\hbar}{2}\left|\pm\hat{x}\right\rangle$,

$$\left|\tilde{\psi}(t)\right\rangle \simeq \frac{1}{\sqrt{2}}\left(e^{-i\Omega t/4}\left|+\hat{x}\right\rangle + e^{i\Omega t/4}\left|-\hat{x}\right\rangle\right)$$

$$\left|\tilde{\psi}(t)\right\rangle \xrightarrow{S_x} \frac{1}{\sqrt{2}}\begin{pmatrix} e^{-i\Omega t/4} \\ e^{i\Omega t/4}\end{pmatrix}.$$

Since U_0 is a function of S_z, $[U_0, S_z] = 0$, so

$$\langle S_z\rangle = \left\langle\psi(t)\right| S_z \left|\psi(t)\right\rangle = \left\langle\tilde{\psi}(t)\right| S_z \left|\tilde{\psi}(t)\right\rangle.$$

With $S_z \xrightarrow{S_x} \frac{\hbar}{2}\sigma_x$ (problem "$\hat{S}_z$ in the S_x basis"),

$$\langle S_z\rangle = \frac{1}{2}\begin{pmatrix} e^{i\Omega t/4} & e^{-i\Omega t/4}\end{pmatrix}\frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix} e^{-i\Omega t/4} \\ e^{i\Omega t/4}\end{pmatrix} = \frac{\hbar}{4}\left(e^{i\Omega t/2} + e^{-i\Omega t/2}\right).$$

Result 26 (Rabi oscillations)**.**

$$\langle S_z(t)\rangle = \frac{\hbar}{2}\cos\frac{\Omega t}{2}, \qquad P_{+\hat{z}}(t) = \cos^2\frac{\Omega t}{4}. \tag{86}$$

The spin flips back and forth between $|+\hat{z}\rangle$ and $|-\hat{z}\rangle$. A drive left on for $\Omega t/2 = \pi$ (a "π-pulse") takes $|+\hat{z}\rangle$ to $|-\hat{z}\rangle$: this is how a spin qubit is flipped.

Remark 52 (WARNING: factor of 2)**.** *Bill's notes write the phases as $e^{\mp i\Omega t/2}$ and get $\langle S_z\rangle = \frac{\hbar}{2}\cos\Omega t$. The eigenvalues of S_x are $\pm\hbar/2$, not $\pm\hbar$, so the phases are $e^{\mp i\Omega t/4}$ and the Rabi frequency is $\Omega/2$. A numerical solution of the full $H(t)$ confirms $\cos(\Omega t/2)$. The half comes from the RWA: $\cos\omega t$ splits into two counter-rotating pieces and only one of them is kept.*

Problem 44 (Rabi formula (PS4 Q10))**.** *Derive the Rabi formula [Townsend, Eq. (4.45)].*

9 Misc

Energy-Time uncertainty: $\Delta E\Delta t \geq \hbar/2$. Recall $\Delta A\Delta B \geq \frac{1}{2}|\langle[A,B]\rangle|$ for any two operators A and B. Take $A = J_x, B = J_y$ and $[J_x, J_y] = i\hbar J_z$. Then $\Delta J_x \Delta J_y \geq \frac{\hbar}{2}|\langle J_z\rangle|$. $\langle H\rangle = E$. Let observable A (not a constant of the motion), such that $[H,A] \neq 0$. $\frac{\Delta A}{\Delta t} \approx |\langle\frac{dA(t)}{dt}\rangle|\,|\langle A(t)\rangle = \langle\psi(t)|\,A\,|\psi(t)\rangle = \langle\psi(0)|\,\dagger\, U(t)AU(t)\,|\psi(0)\rangle = \langle\psi(0)|\,A\,|\psi(0)\rangle$. $\implies \Delta A = |\langle\frac{dA(t)}{dt}\rangle|\Delta t$, then $\Delta E\Delta A \geq 1/2|\langle[A,H]\rangle| = \frac{\hbar}{2}|\langle\frac{dA(t)}{dt}\rangle|$. $\implies \Delta E\Delta t \geq \hbar/2$.

Chapter 5: two spins $1/2$. complete set $\{|+\hat{z},+\hat{z}\rangle, |+\hat{z},-\hat{z}\rangle, |-\hat{z},+\hat{z}\rangle, |-\hat{z},-\hat{z}\rangle\}$, S_{z1} and S_{z2} are just the first and second item in the kets, and $[S_{z1}, S_{z2}] = 0$. Or: $\{|+\hat{z},+\hat{x}\rangle, |+\hat{z},-\hat{x}\rangle, |-\hat{z},+\hat{x}\rangle, |-\hat{z},-\hat{x}\rangle\}$. $|+\hat{z},+\hat{z}\rangle = |+\hat{z}\rangle\otimes|+\hat{z}\rangle = |+\hat{z}\rangle_1|+\hat{z}\rangle_2$. $S_{z2}|+\hat{z}\rangle\otimes|+\hat{z}\rangle = |+\hat{z}\rangle\otimes S_{z2}|+\hat{z}\rangle = \frac{\hbar}{2}(|+\hat{z}\rangle\otimes|+\hat{z}\rangle)$. $\{|1\rangle, |2\rangle, |3\rangle, |4\rangle\}$, where $|1\rangle = |+\hat{z},+\hat{z}\rangle$, $|2\rangle = |+\hat{z},-\hat{z}\rangle$, $|3\rangle = |-\hat{z},+\hat{z}\rangle$, $|4\rangle = |-\hat{z},-\hat{z}\rangle$. $\vec{S_2}\cdot\vec{S_2} = S_{1z}S_{2z} - 1/2(S_{1+}S_{2-} + S_{1-}S_{2+})$. $S_{i\pm} = S_{ix} \pm iS_{iy}$, $H = \frac{A}{\hbar^2}\vec{S_i}\vec{S_2} \overset{|1_j\rangle-basis}{\rightarrow}$

$$\begin{pmatrix} \langle 1|\,H\,|1\rangle & \langle 1|\,H\,|2\rangle & \langle 1|\,H\,|3\rangle & \langle 1|\,H\,|4\rangle \\ \langle 2|\,H\,|1\rangle & \langle 2|\,H\,|2\rangle & \langle 2|\,H\,|3\rangle & \langle 2|\,H\,|4\rangle \\ \langle 3|\,H\,|1\rangle & \langle 3|\,H\,|2\rangle & \langle 3|\,H\,|3\rangle & \langle 3|\,H\,|4\rangle \\ \langle 4|\,H\,|1\rangle & \langle 4|\,H\,|2\rangle & \langle 4|\,H\,|3\rangle & \langle 4|\,H\,|4\rangle \end{pmatrix}.$$

A Solutions

B Useful Links

Stern-Gerlach experiment:
https://www.youtube.com/shorts/QG7kJXxN1Kk